

Rigid Analytic Geometry

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1 Introduction - Spectral Spaces and Spaces of Valuation Spectra

1.1 Introduction

Definition 1.1. Let $\Omega \subseteq \mathbb{C}$ be a domain. We define a sheaf of holomorphic functions as,

$$\Omega \supseteq V \mapsto \mathcal{O}(V) = \{f : V \rightarrow \mathbb{C} \mid f \text{ is holomorphic}\}.$$

One can equip $\mathcal{O}(V)$ this with the semi-norm such that,

$$\|f\|_{V,A} = \max\{|f(z)| \mid z \in A\},$$

where $A \subseteq \Omega$ is a closed and bounded in $\mathbb{C}$, namely a compact set A . With respect to this semi-norm, we define basis elements of the topology in the following way.

Proposition 1.1. The semi-norm induces the topology on $\mathcal{O}(V)$ whose basis elements of the neighbourhood of 0 are of the following form. For $f \in \mathcal{O}(V)$ and $\varepsilon > 0$,

$$\{f \in \mathcal{O}(V) \mid \|f\|_{V,A} < \varepsilon\}.$$

Definition 1.2. A ring is said to be Bézout if every finitely generated ideal is principal.

Remark 1.1. By Weierstrass product theorem, $\mathcal{O}(V)$ is not Noetherian but it is Bézout.

Proposition 1.2. The stalks of $\mathcal{O}(V)$ which is a ring of germs of holomorphic functions are Noetherian. This holds for holomorphic functions in several complex variables as well.

The goal of the course is to generalize these results to p -adic cases.

We will allow ground field K with an absolute value $|\cdot| : K \rightarrow [0, \infty)$ satisfying,

$$|0| = 0, |1| = 1, |ab| = |a||b|, |a + b| \leq \max\{|a|, |b|\}.$$

Definition 1.3. Let K be a field which is complete with respect to the map $|\cdot|_K : K \rightarrow [0, \infty)$ is an absolute value of K if we have the following conditions.

- i). $|1|_K = 1$,
- ii). $|0|_K = 0$,
- iii). $\forall a, b \in K, |ab|_K = |a|_K |b|_K$.

Furthermore, it is called ultra-metric if we have,

$$\forall a, b, |a + b|_K \leq \max\{|a|_K, |b|_K\}.$$

Proposition 1.3. Suppose we have an ultra-metric $\text{abs} \cdot_K$ on a field K . If K is complete with respect to the topology induced by the metric $d(x, y) = |x - y|_K$, then for a finite extension L/K , there is an extension $|\cdot|_L : L \rightarrow [0, \infty)$ of $|\cdot|_K$ such that L is complete with respect to the canonical metric obtained from $|\cdot|_L$.

Proposition 1.4. Let K be a field with an ultrametric $|\cdot|_K$. The completion $\widehat{\overline{K}}$ of the algebraic closure $\overline{K}$ of K with respect to $|\cdot|_K$ is again algebraically closed.

Example 1.1. In Proposition 1.4, we can take for example $\widehat{\overline{K}}$ to be $\mathbb{C}_p$ where $K = \mathbb{Q}_p$.

We will now give the definition of our basic objects.

Definition 1.4. Let $(K, |\cdot|_K)$ be a field which is complete with respect to the absolute value $|\cdot|_K$. We define the Tate algebra over K as

$$\mathbb{T}_m(K) := \left\{ f = \sum_{\alpha \in \mathbb{N}^m} f_\alpha X^\alpha \mid \begin{array}{l} \text{for all } \varepsilon > 0, \text{ there are only} \\ \text{finitely many } \alpha \text{ such that } |f_\alpha| > \varepsilon \end{array} \right\}$$

We define,

$$\|f\|_{\mathbb{T}_m} := \max\{|f_\alpha|_K \mid \alpha \in \mathbb{N}^m\}.$$

Proposition 1.5. Suppose K is algebraically closed in Definition 1.4. Then we have,

$$\|f\|_{\mathbb{T}_m} = \max\{|f(z)| \mid z = (z_i)_{i=1, \dots, m} \in K^m, |z_i| \leq 1\},$$

and such maximal is actually attained.

Remark 1.2. In p -adic settings, K^m is totally disconnected so assertion like the "identity theorem" for holomorphic functions becomes problematic.

In order to resolve the above obstacle Grothendieck introduced Étale topology.

1.2 Main Theorems of the Course

Definition 1.5. An element a is power-bounded with respect to a norm $\|\cdot\|$ if there is $M > 0$ such that for all $n \in \mathbb{N}$, $\|a\|^n \leq M$.

Definition 1.6. Let A be a topological ring. An element $a \in A$ is topologically nilpotent if

$$\lim_{n \rightarrow \infty} a^n = 0.$$

Definition 1.7. We define the following subset of a Tate algebra $\mathbb{T}_m$.

$$\begin{aligned} \mathbb{T}_m^\circ &:= \{f \in \mathbb{T}_m \mid f \text{ is power bounded}\}, \\ \mathbb{T}_m^{\circ\circ} &:= \{f \in \mathbb{T}_m \mid f \text{ is topologically nilpotent } \lim_{k \rightarrow \infty} f^k = 0\}. \end{aligned}$$

Proposition 1.6. $\|\cdot\|_{\mathbb{T}_m}$ is multiplicative. (i.e. $\|fg\| = \|f\|\|g\|$ and $\|1\| = 1$). In particular,

$$\mathbb{T}_m^\circ = \{f \in \mathbb{T}_m(K) \mid \|f\|_{\mathbb{T}_m} \leq 1\}.$$

and

$$\mathbb{T}_m^{\circ\circ} = \{f \in \mathbb{T}_m(K) \mid \|f\|_{\mathbb{T}_m} < 1\}.$$

From the definition of ultrametric, it is clear that $\mathbb{T}_m^\circ$ is a subring and $\mathbb{T}_m^{\circ\circ}$ is an ideal.

Proposition 1.7. $\mathbb{T}_m(K)$ is Noetherian and all ideals are closed in it. If $\mathfrak{m}$ is a maximal ideal, its residue field $k(\mathfrak{m}) = \mathbb{T}_m/\mathfrak{m}$ is a finite field extension of K (thus $|\cdot|_K$ extends uniquely to $k(\mathfrak{m})$).

Proposition 1.8. There is a topologically nilpotent unit, thus there are no proper open ideals.

Proposition 1.9. $\mathbb{T}_m(K)$ is regular of Krull dimension m .

Definition 1.8. A ring A is Jacobson if every prime ideal is an intersection of some maximal ideals.

Proposition 1.10. $\mathbb{T}_m(K)$ is a Jacobson ring. (The closed points of $\text{Spec } \mathbb{T}_m$ are dense in every closed subsets)

Notation 1.1. For a ring A , we denote

$$\text{Sp } A := \{\mathfrak{m} \in \text{Spec } A \mid \mathfrak{m} \text{ is a maximal ideal}\}.$$

Notation 1.2. Let $f \in \mathbb{T}_m(K)$ and $x \in \text{Sp } \mathbb{T}_m(K)$. We denote,

$$f(x) := f \bmod x \in k(x).$$

Proposition 1.11.

$$\|f\|_{\mathbb{T}_m} = \max\{|f(x)| \mid x \in \text{Sp } \mathbb{T}_m\}.$$

We define a basis element of the topology in $\mathbb{T}_m(K)$ as follows.

Definition 1.9 (Rational Open Sets). *Suppose $\mathbb{T}_m(K) = \langle f_0, \dots, f_m \rangle_{\mathbb{T}_m}$. The rational open set associated to the generators $\langle f_0, \dots, f_m \rangle_{\mathbb{T}_m}$ is*

$$\mathcal{R}_{\text{Sp } \mathbb{T}_m}(f_0|f_1, \dots, f_m) = \{x \in \text{Sp } \mathbb{T}_m \mid |f_0(x)| \geq |f_i(x)|, 1 \leq i \leq m\}.$$

Remark 1.3. *The Grothendieck topology enforces the quasi-compactness of these subsets.*

Additional points were introduced by van der Put in the late 1970s using a filter machinery equivalent to topos points. He also gave equivalence with valuation points not

Berkovich in the 1980s constructed a multiplicative seminorm $\mathbb{T}_m \rightarrow [0, \infty)$ which are similar to $f \mapsto |f(x)|$ for $x \in \mathbb{T}_m$ or $f \mapsto \|f|_{\mathbb{T}_m}\|$. One then gets a compact Hausdorff space.

Huber constructed more general valuations and one gets a spectral space with a sheaf category equivalent to sheaves on $\text{Sp } \mathbb{T}_m$ (unlike Berkovich).

In the Berkovich machinery, one also has a point corresponding to

$$\|f\|_r = \max\{r^{|\alpha|} |f_\alpha|_K \mid \alpha \in \mathbb{N}^m\},$$

where $|\alpha| = \sum \alpha_i$ and $0 \leq r \leq 1$.

Remark 1.4.

$$\|f\|_1 = \|f|_{\mathbb{T}_m}\|, \|f\|_0 = 0.$$

Definition 1.10. *Let $0 \leq r \leq 1$. We define, for $f \in \mathbb{T}_m(K) \setminus \{0\}$,*

$$N_r(f) := (\|f\|_r, n)$$

where $n = \max\{i \in \mathbb{N} \mid \|f\|_r = r^i |f_i|_K\}$.

Example 1.2. *Consider the case $\mathbb{T}_1(K)$ and $f \in \mathbb{T}_1(K) \setminus \{0\}$ where*

$$f = \sum_{n=0}^{\infty} f_n T^n.$$

Then N_r defines a valuation, $\mathbb{T}_1 \rightarrow \Gamma \cup \{0\}$ where $\Gamma = (0, \infty)_{\mathbb{R}} \times \mathbb{Z}$ ordered lexicographically with the first factor being the most significant and $0 < \gamma$ for all $\gamma \in \Gamma$.

One has

$$N(a, b) = N(a)N(b), N(1) = 1, N(0) = 0, N(a + b) \leq \max\{N(a), N(b)\}. \quad (1.1)$$

When $0 < r < 1$ and $f \in \mathbb{T}_1 \setminus \{0\}$, there is $R \in (r, 1]_{\mathbb{R}}$ such that

$$f(z) = \|f_r\| \left(\frac{|z|}{r} \right)^{n'}, r < |z| \leq R. \quad (1.2)$$

where N is as in Equation (1.1), and $\|f\|_r = \sup\{|f(z)| \mid |z| \leq r\}$ where $r = 1$, Equation (1.2) has no analog as no R extends as $f(z)$ is undefined where $|z| > 1$. This means that N_1 is extended

Definition 1.11.

$$\text{Spa}(\mathbb{T}_m, \mathbb{T}_m^0) = \{\text{continuous valuations } \gamma \text{ of } \mathbb{T}_m \text{ s.t. } \forall f \in \mathbb{T}_m^\circ, \gamma(f) \leq 1\}.$$

Remark 1.5. $N_1(T) = (1, 1)$, $N_1(T^{2026}) = (1, 2026)$.

1.3 Spectral Spaces

Remark 1.6. $\emptyset$ is not irreducible.

Proposition 1.12. Let X be a topological space. For a point $x \in X$,

$$\overline{\{x\}} = \{y \in X \mid \text{every neighbourhood of } y \text{ in } X \text{ contains } x\}.$$

Definition 1.12. A point $x \in X$ is a generic point if $\overline{\{x\}} = X$, (i.e. $\{x\}$ is dense in X).

Notation 1.3. For a topological space X , we denote the collection of all open sets to be $\mathcal{O}_X$.

Definition 1.13. Let X be a topological space and $x, y \in X$. We say x, y are topologically distinguishable if there is $U \in \mathcal{O}_X$ such that either $x \in U$ and $y \notin U$ or $x \notin U$ and $y \in U$. They are otherwise indistinguishable.

Definition 1.14. A topological space X is T_0 or Kolmogorov if for any pair of points are distinguishable.

Definition 1.15. A topological space X is sober if it satisfies one of the following equivalent conditions.

- 1). Every irreducible neighborhood of X has precisely one generic point.
- 2). X is T_0 and every irreducible subset has at least one generic point.

Remark 1.7. It is easy to see that these are local conditions.

Definition 1.16. A topological space is T_1 if it satisfies one of the following equivalent conditions.

- 1). For any $x, y \in X$, $x \neq y \Rightarrow$ there is $U \in \mathcal{O}_X$ such that $x \in U$ and $y \notin U$.
- 2). Every point of X is closed.

It is T_2 or Hausdorff if it satisfies one of the following equivalent conditions.

- 1). Any pair of points $x, y \in X$ has disjoint neighborhoods U, V such that $x \in U$ and $y \in V$.
- 2). The diagonal set $\{(x, x) \mid x \in X\}$ is closed in $X \times X$.

Definition 1.17. X is quasi-compact if every open covering admits a finite subcovering. A subset M of X is quasi-compact if it is a quasi-compact as a topological space with the subset topology from X .

Proposition 1.13. *If X can be covered by finitely many quasi-compact open neighbourhoods then it is quasi-compact.*

Proposition 1.14. *Every closed subset of a quasi-compact space is again quasi-compact.*

Proposition 1.15. *Let $X \xrightarrow{f} Y$ be a continuous with X being quasi-compact and Y being Hausdorff, then f is closed (i.e. every closed set A of X has a closed image $f(A)$ in Y).*

Definition 1.18. *A subset $\mathcal{B} \subseteq \mathcal{O}_X$ is a topology basis for X if for any $U \in \mathcal{O}_X$, we have $U = \bigcup_{V \in \mathcal{B}_U} V$ where $\mathcal{B}_U = \{V \in \mathcal{B} \mid V \subseteq U\}$.*

Remark 1.8. *The subsets of the form*

$$D(f) = \{\mathfrak{p} \in \text{Spec } A \mid f \notin \mathfrak{p}\}$$

form a topology basis for $\text{Spec } A$.

Proposition 1.16. *For a topological space X , the following conditions are equivalent.*

- i). $\mathfrak{Q}_C(X) := \{\Omega \in \mathcal{O}_X \mid \Omega \text{ is quasi-compact}\}$ is a topology basis closed under finite intersection (including the empty intersection $\bigcap_{\emptyset} = X$).
- ii). *There is a topology base $\mathcal{B} \subseteq \mathfrak{Q}_C(X)$ such that $\mathcal{B}$ consists of quasi-compact and it is closed under the finite intersection in X .*

Definition 1.19. *A topological space X is spectral if it is sober and satisfies one of the conditions in Proposition 1.16.*

Definition 1.20. *A spectral map is a map $X \xrightarrow{f} Y$ between spectral spaces such that for any quasi-compact subset Ω in Y , $f^{-1}(\Omega)$ is quasi-compact subset of X .*

Remark 1.9. *A spectral map is automatically continuous.*

Proposition 1.17. *If $\mathcal{B} \subseteq \mathcal{O}_Y$ is a topology base as in the second condition of Proposition 1.16 for Y , then f is spectral if and only if $\forall \Omega \in \mathcal{B}, f^{-1}(\Omega) \in \mathfrak{Q}_C(X)$.*

Example 1.3. *$\text{Spec } A$ is spectral and every ring homomorphism defines a spectral map and $\{D(f) \mid f \in A\}$ are such base as in Proposition 1.16 for $\text{Spec } A$.*

Example 1.4. *The underlying space of qcqs (i.e. quasi-compact and quasi-separated) scheme is spectral.*

Remark 1.10. *Every closed and every quasi-compact open sets of a spectral space is spectral.*

Remark 1.11. *A Noetherian space X (i.e. $\mathfrak{Q}_C(X) = \mathcal{O}_X$) is spectral if and only if it is sober.*

Proposition 1.18. *Every non-empty closed subset of a spectral space X contains a closed point.*

Proof. By Remark 1.10, it suffices to show that every spectral space $X \neq \emptyset$ contains a closed point as non-empty closed subsets are again spectral. Set,

$$\mathcal{M} := \{Z \subseteq X \mid Z \text{ is closed}\}.$$

Equip $\mathcal{M}$ with a partial order by $\subseteq$. Since X is quasi-compact, $\mathcal{M}$ has a minimal element with such order. Let Z be a such minimal element in $\mathcal{M}$. If Z has two different element x, y , then by T_0 condition, we have $U \in \mathcal{O}_X$ such that $x \notin U$ and $y \in U$. Then $Z' = Z \setminus U$ is in $\mathcal{M}$. As $x \in Z'$, this contradicts to the minimality of Z as $y \in Z \setminus Z'$. $\square$

Definition 1.21. *A filter base of a set M is a non-empty subset $\mathcal{B}$ of $\mathfrak{P}(M)$ such that*

- i). $\emptyset \notin \mathcal{B}$,
- ii). for $N, O \in \mathcal{B}$ there is $P \in \mathcal{B}$ with $P \subseteq N \cap O$.

A filter is a filter base $\mathcal{F}$ such that for $N \in \mathcal{F}$, and $N \subseteq O \subseteq M$ implies $O \in \mathcal{F}$.

A filter generated by a filter base $\mathcal{B}$ is such that

$$\mathcal{F} := \{N \mid N \subseteq M \text{ and there is } O \in \mathcal{B} \text{ such that } O \subseteq N\}.$$

A direct image under a map $M \xrightarrow{f} X$ of a filter $\mathcal{F}$ is such that

$$f_*\mathcal{F} := \{Y \mid Y \subseteq X \text{ and } f^{-1}(Y) \in \mathcal{F}\}.$$

Example 1.5. *A neighbourhood filter $\mathcal{N}_x$ of $x \in X$ is such that*

$$\mathcal{N}_x := \{N \subseteq X \mid \exists U \in \mathcal{O}_X \text{ s.t. } x \in U \subseteq N\}.$$

Example 1.6. *A principal filter $\mathcal{P}_x$ of $x \in X$ is such that*

$$\mathcal{P}_x = \{N \subseteq X \mid x \in N\}.$$

Remark 1.12. *A neighbourhood base for $x \in X$ is a filter base $\mathcal{B}$ such that it generates the neighbourhood filter of x .*

Remark 1.13. *The notion of filters is introduced by Cartan.*

Proposition 1.19. *For a filter $\mathcal{F}$ of X , the following statements are equivalent.*

- 1). $\mathcal{F}$ is a $\subseteq$ -maximal element of the set of filters on X .
- 2). If $M, N \subseteq X$ and $M \cup N \in \mathcal{F}$ then $M \in \mathcal{F}$ or $N \in \mathcal{F}$.
- 3). For $M \subseteq X$, precisely one of M and $X \setminus M$ is in $\mathcal{F}$.

Definition 1.22. Such filter $\mathcal{F}$ satisfying the conditions in Proposition 1.19 is called an *ultrafilter*.

Theorem 1.1 (Tarski). *Every filter on X is contained in some ultrafilter.*

Proof. Follows from Zorn's lemma. $\square$

Proposition 1.20. *Let $\mathcal{U}$ be an ultrafilter of X then all direct images of $\mathcal{U}$ are ultrafilters. If $\iota : A \rightarrow X$ is the inclusion of a subset then this induces,*

$$\{\text{ultrafilters on } A\} \xrightarrow{\iota_*} \{\text{ultrafilters on } X \text{ containing } A\},$$

which is a bijection.

Definition 1.23.

$$\text{Lim } \mathcal{F} = \{x \in X \mid \mathcal{N}_x \subseteq \mathcal{F}\}.$$

is the set of limits of $\mathcal{F}$. If it contains precisely one $x \in X$, we denote it by

$$\text{Lim } \mathcal{F} = x.$$

Definition 1.24. *We say that $x \in X$ is a point of accumulation (denoted by point of accumulation, sometimes called point of condensation) if for every neighbourhood $V \in \mathcal{N}_x$ and every $M \in \mathcal{F}$, we have $V \cap M \neq \emptyset$.*

Proposition 1.21. *Let $x \in X$ then*

$$\overline{\{x\}} = \text{Lim } \mathcal{P}_x = \text{point of accumulation of the principal filter of } x = \text{Lim } \mathcal{N}_x.$$

Proposition 1.22. *Let $\mathcal{F}, \mathcal{G}$ be filters such that $\mathcal{F} \subseteq \mathcal{G}$ and $\mathcal{U}$ be a ultrafilter.*

1. *Every limit of $\mathcal{F}$ is a point of accumulation of $\mathcal{F}$.*
2. *Every limit of $\mathcal{F}$ is a limit of $\mathcal{G}$.*
3. *Every point of accumulation of $\mathcal{G}$ is a point of accumulation of $\mathcal{F}$.*
4. *A point $x \in X$ is a limit of $\mathcal{U}$ if and only if it is a point of accumulation of $\mathcal{U}$.*
5. *The sets of its limits and its point of accumulations are closed for $\mathcal{F}$.*
6. *Let $\iota : A \rightarrow X$ be a canonical inclusion where A has an induced topology from X . Then $\text{Lim } \mathcal{F} = A \cap \text{Lim } \iota_* \mathcal{F}$.*

Proof. For the third assertion, if a is a point of accumulation of $\mathcal{U}$ and $V \in \mathcal{N}_x$, then $X \setminus V \notin \mathcal{U}$ as x is a point of accumulation. Hence $V \in \mathcal{U}$ by Proposition 1.19. For the forth assertion, let $x \notin \text{Lim } \mathcal{F}$, (respectively x not a point of accumulation of $\mathcal{F}$) then there is $U \in \mathcal{N}_x$ such that $U \notin \mathcal{F}$ (respectively, such that there is $M \in \mathcal{F}$ such that $U \cap M = \emptyset$). Then no element of U is a limit (respectively, point of accumulation) of $\mathcal{F}$. $\square$

Proposition 1.23. *For a subset M of a topological space X , the following subsets all coincide.*

- a). $\{\text{limits of ultrafilters } \mathcal{U} \text{ of } X \text{ such that } M \in \mathcal{U}\},$
- b). $\bigcup_{M \in \mathcal{F}}^{\text{filter}} \text{Lim } \mathcal{F}.$
- c). $\{x \in X \mid \text{there is a filter } \mathcal{F} \text{ such that } M \in \mathcal{F} \text{ and } x \text{ is a point of accumulation of } \mathcal{F}\}.$
- d). $\bigcap_{\substack{M \subseteq Z \subseteq X \\ Z \text{ closed}}} Z.$

Proof. Denote those sets by M_a, M_b, M_c, M_d . Then,

$$M_a \underbrace{\subseteq}_{\text{trivial}} M_b \underbrace{\subseteq}_{\text{Proposition 1.22}} M_c \underbrace{\subseteq}_{\text{definition}} M_d.$$

For the reverse direction, note that $M_d = \bigcup_{\substack{U \in \mathcal{O}_X, \\ U \cap M = \emptyset}} (X \setminus U)$. If $x \in M_d$, then

$$\{U \cap M \mid U \in \mathcal{N}_x\}$$

is a filter base containing M and in particular containing $\mathcal{N}_x$. By Theorem 1.1, it is contained in some ultrafilter $\mathcal{U}$ and by definition $x \in \text{Lim } \mathcal{U}$. $\square$

Definition 1.25. *The set of above is called the closure of M in X .*

Remark 1.14. *The closure of a point s is a special case of this.*

Corollary 1.1. *For a subset A of a topological space X , the following conditions are equivalent.*

- i). *If $\mathcal{F}$ is a filter on X such that $A \in \mathcal{F}$ and x is a limit of $\mathcal{F}$ then $x \in A$.*
- ii). *If $\mathcal{F}$ is a filter on X such that $A \in \mathcal{F}$ and x is a point of accumulation of $\mathcal{F}$ then $x \in A$.*
- iii). *If $\mathcal{U}$ is a ultrafilter on X such that $A \in \mathcal{U}$, every limit of $\mathcal{U}$ in X is in A .*
- iv). *A is closed.*

Proof. Clear from Proposition 1.23 $\square$

Proposition 1.24. *For a map $f : X \rightarrow Y$ of topological spaces. The following conditions are equivalent.*

- i). *f is continuous.*
- ii). *If $\mathcal{F}$ is a filter on X and $x \in \text{Lim } \mathcal{F}$ then $f(x) \in \text{Lim } f_* \mathcal{F}$.*
- iii). *If $\mathcal{F}$ is a filter on X and $x \in X$ is a point of accumulation of $\mathcal{F}$, then $f(x)$ is a point of accumulation of $f_* \mathcal{F}$.*
- iv). *If $\mathcal{U}$ is a ultrafilter on X and $x \in \text{Lim } \mathcal{U}$, then $f(x) \in \text{Lim } f_* \mathcal{U}$.*

Proof.

- i) $\Rightarrow$ ii). This follows from that $f_*\mathcal{N}_x = \mathcal{N}_{f(x)}$ and f_* preserves inclusion.
- ii) $\Rightarrow$ iii). By Proposition 1.22, every limit is a point of accumulation.
- iii) $\Rightarrow$ iv). By Proposition 1.22, every point of accumulation of an ultrafilter is a limit of the filter and f_* maps ultrafilters to ultrafilters.
- i) $\Rightarrow$ iv). If $A \subseteq Y$ is closed, we want to show that $f^{-1}(A)$ is closed. Let $\mathcal{U}$ be an ultrafilter in X such that $f^{-1}(A) \in \mathcal{U}$. Then $f_*\mathcal{U}$ is an ultrafilter on Y containing A and by Corollary 1.1, any limit of $f_*\mathcal{U}$ is in A . Since limits of $\mathcal{U}$ are mapped to limits of $f_*\mathcal{U}$, $f^{-1}(A)$ contains limits of $\mathcal{U}$. Again by Corollary 1.1 as $f^{-1}(A)$ is closed. $\square$

Remark 1.15. *By Definition 1.24, we have the following. The set of points of accumulation of a filter $\mathcal{F}$ is the intersection of the closure of the elements of $\mathcal{F}$ or the intersection of all $A \in \mathcal{F}$ which are closed in X .*

Proposition 1.25. *Let X be a topological space. The following conditions are equivalent.*

- i). X is Hausdorff.
- ii). Every filter has at most one limit.
- iii). Every ultrafilter on X has at most one limit.

Proof.

i) $\Rightarrow$ ii). If $x \neq y$, they have disjoint neighborhoods $x \in U$ and $y \in V$. If $U \in \mathcal{F}$ and $V \in \mathcal{F}$, then $\emptyset = U \cap V \in \mathcal{F}$, which is not possible. Thus precisely one of x and y is in $\text{Lim } \mathcal{F}$.

ii) $\Rightarrow$ iii) is trivial.

iii) $\Rightarrow$ i). Assume $x \neq y$ and consider the set

$$\{U \cap V \mid U \in \mathcal{N}_x, V \in \mathcal{N}_y\}.$$

This is a filter unless it contains $\emptyset$. In that case, it is contained in some ultrafilter $\mathcal{U}$. By definition of limits, x, y are both limits of $\mathcal{U}$ thus $x = y$ which is a contradiction. $\square$

Proposition 1.26. *Let X be a topological space. The following conditions are equivalent.*

- i'). X is quasi-compact.
- ii'). Every filter has at least one point of accumulation.
- iii'). Every ultrafilter has at least one limit.

Proof.

i') $\Rightarrow$ ii') By Remark 1.15,

$$\{x \mid x \text{ is a point of accumulation of } \mathcal{F}\} = \bigcap_{M \in \mathcal{F}} \overline{M}.$$

By quasi-compactness X and applying it to the complement, this is a finite intersection of elements in $\mathcal{F}$. By definition, any filter is closed under finite intersection and does not contain the emptyset, this is not empty.

ii') $\Rightarrow$ iii'). Every filter is contained in some ultrafilter and limits and points of accumulation are the same in ultrafilter by Proposition 1.22.

iii') $\Rightarrow$ i'). Let $(A_\lambda)_\Lambda$ be a family of closed subsets such that for any finite subset $F \subseteq \Lambda$, we have,

$$\bigcap_{\lambda \in F} A_\lambda \neq \emptyset.$$

Then the collection,

$$\left\{ \bigcap_{\lambda \in F} A_\lambda \mid F \subseteq \Lambda, |F| < \infty \right\}$$

is a filter base, thus it is contained in some ultrafilter $\mathcal{U}$. Let $x \in \text{Lim}\mathcal{U}$. For $\lambda \in \Lambda$, $A_\lambda \in \mathcal{U}$ hence $x \in A_\lambda$ by Corollary 1.1. Hence $x \in \bigcap_{\lambda \in \Lambda} A_\lambda$ and this intersection is not empty. This is the definition of quasi-compactness stated in terms of closed sets. $\square$

Proposition 1.27. *If X satisfies these equivalent conditions of Proposition 1.16, then for every ultrafilter $\mathcal{U}$ on X , the closed subset $\text{Lim}\mathcal{U} \subseteq X$ is irreducible.*

Proof. Let $\Omega, \Theta \in \mathfrak{Q}_C(X)$ such that $\Omega \cap \text{Lim}\mathcal{U} \neq \emptyset$ and $\Theta \cap \text{Lim}\mathcal{U} \neq \emptyset$. By the definition of $\text{Lim}\mathcal{U}$, $\Omega, \Theta \in \mathcal{U}$. Hence $\Omega \cap \Theta \in \mathcal{U}$. As $\Omega \cap \Theta$ is quasi-compact $\mathcal{U} \cap \mathcal{O}_{\Omega \cap \Theta}$ has a limit $x \in \Omega \cap \Theta$ by Proposition 1.26. By Proposition 1.22, $x \in \text{Lim}\mathcal{U}$.

If U and V are open subsets such that $U \cap \text{Lim}\mathcal{U}$ and $V \cap \text{Lim}\mathcal{U}$ are not emptysets, there are $\Omega \subseteq U$ and $\Theta \subseteq V$ as above (as $\mathfrak{Q}_C(X)$ is a topology base), hence $U \cap V \cap \text{Lim}\mathcal{U}$ is not empty. Finally by Proposition 1.26, $\text{Lim}\mathcal{U} \neq \emptyset$ as X is quasi-compact. $\square$

Definition 1.26. *We say that $x \in X$ is a generic limit of a ultrafilter $\mathcal{U}$ if $\text{Lim}\mathcal{U} = \overline{\{x\}}$.*

Definition 1.27. *Let X be a topological space in which every ultrafilter $\mathcal{U}$ has a unique generic limit. Then we write,*

$$\text{glim}\mathcal{U} = \text{glim}_X \mathcal{U}$$

We say that a subset $M \subseteq X$ is closed under generic limit in X if $M \in \mathcal{U}$ for every ultrafilter $\mathcal{U}$ implies $\text{glim}\mathcal{U} \in M$.

Proposition 1.28.

- 1). X is T_0 if and only if every ultrafilter on X has at most one generic limit.
- 2). If every ultrafilter on X has some generic points then every closed subset $Z \subseteq X$ has some generic points.

Proof. For 1). the sufficiency of being T_0 is by Definition 1.15. If x and y are topologically indistinguishable, then $\overline{\{x\}} = \overline{\{y\}} = Z$ for some Z . Furthermore, the principal filter $\mathcal{U}$ of x which has $\text{Lim}\mathcal{U} = Z$ from Example 1.6 has both x and y as generic limits.

2). Irreducibility of Z implies that $\mathcal{O}_Z \setminus \{\emptyset\}$ is a filter base in X . Hence it is contained in some ultrafilter $\mathcal{U}$. By Definition 1.25 and the closedness of Z implies $Z = \text{lim}\mathcal{U}$. Hence every $\text{glim}\mathcal{U}$ is a generic point of Z . $\square$

Proposition 1.29. *Assume every ultrafilter $\mathcal{U}$ on X has a unique generic limit $\text{glim}\mathcal{U}$. Then the following statements hold.*

- 1). *The set of subsets of X which are closed under glim is closed under arbitrary intersection and finite unions.*
- 2). *An open subset $V \subseteq X$ is closed under glim if and only if V is quasi-compact.*

Proof. For 1), the closedness under the intersections is obvious. If $M \cup N \in \mathcal{U}$ then $M \in \mathcal{U}$ or $N \in \mathcal{U}$ by Proposition 1.19. Hence $\text{glim}\mathcal{U} \in M \subseteq M \cup N$ or $\text{glim}\mathcal{U} \in N \subseteq M \cup N$.

Let $\mathcal{V}$ be an ultrafilter on V and $\mathcal{U}$ be the image of $\mathcal{V}$ under the embedding $V \hookrightarrow X$. If V is closed under glim then $\text{glim}\mathcal{V} \in V$ is a limit of $\mathcal{U}$ by Proposition 1.22. By Proposition 1.25, V is quasi-compact. If V is quasi-compact, $\mathcal{V}$ has a limit $x \in V$ and $x \in \text{Lim}\mathcal{V} = V \cap \text{Lim}\mathcal{U}$ by Proposition 1.22. Hence $x \in \{y\}$ hence $y \in V$ and V is closed under glim . $\square$

Definition 1.28. *A spectral space X is quasi-separated if for any ultrafilter $\mathcal{U}$ on X , if its limit $\text{Lim}\mathcal{U}$ is non-empty then it is an irreducible closed subset.*

Proposition 1.30.

- 1). *A topological space X is separated if and only if every ultrafilter on X has a unique generic limit and the set of those $\Omega \in \mathcal{O}_X$ which are closed under glim is a topology base.*
- 2). *An open subset of a separated space X is quasi-compact if and only if it is closed under glim . (This is proved in Proposition 1.29).*
- 3). *A map $f : X \rightarrow Y$ is separated if and only if it is continuous and $\text{glim}_Y f_*\mathcal{U} = f(\text{glim}_X \mathcal{U})$ for every ultrafilter $\mathcal{U}$ on X . (This is proved by the above assertion).*

Proof. If X is separated, then by Proposition 1.27, $\text{Lim}\mathcal{U}$ is always closed and irreducible subset, hence $\text{glim}\mathcal{U}$ exists uniquely. By Proposition 1.29, one has $\{U \in \mathcal{O}_X \mid U \text{ is closed under } \text{glim}\} = \mathfrak{Q}_C(X)$ is a topology base. If any ultrafilter has a unique generic limit, then X is sober by Proposition 1.29 and $\mathcal{O}_C(X) = \{U \in \mathcal{O}_X \mid U \text{ is closed under } \text{glim}\}$ which is closed under finite intersection, in X and by the assumption this is a topology base for X . $\square$

Theorem 1.2. *A topological space X is spectral if and only if there is a ring A such that X is homeomorphic to $\text{Spec } A$.*

Proof. We omit the hard part which is $\Rightarrow$ direction (the existence of such ring A for every spectral space X). That $\text{Spec } A$ is spectral can be seen as follows. Let $\mathcal{U}$ be an ultrafilter on $\text{Spec } A$ and set

$$\mathfrak{g}_{\mathcal{U}} := \{a \in A \mid V(a) \in \mathcal{U}\} = \{a \in A \mid \{\mathfrak{p} \in \text{Spec } A \mid a \in \mathfrak{p}\} \in \mathcal{U}\}.$$

We show $\mathfrak{g}_{\mathcal{U}} \in \text{Spec } A$ that is let $a_1, a_2 \in \mathfrak{g}_{\mathcal{U}}$, and $M_i = \{\mathfrak{p} \in \text{Spec } A \mid a_i \in \mathfrak{p}\} \in \mathcal{U}$ hence $M_1 \cap M_2 \in \mathcal{U}$, hence $\{\mathfrak{p} \mid a_1 + a_2 \in \mathfrak{p}\} \in \mathcal{U}$ as $\mathcal{U}$ is closed under finite intersection. Hence $a_1 + a_2 \in \mathfrak{g}_{\mathcal{U}}$ thus it is an ideal. Let $a_1 a_2 \in \mathfrak{g}_{\mathcal{U}}$, but $a_2 \notin \mathfrak{g}_{\mathcal{U}}$. Then $M_2 \notin \mathcal{U}$ but $M_1 \cup M_2 \in \mathcal{U}$ since $a_1 a_2 \in \mathfrak{p}$, one of them must belong to it. Hence

$$M_1 \cup M_2 \supseteq \{\mathfrak{p} \mid a_1 a_2 \in \mathfrak{p}\} \in \mathcal{U}.$$

But $M_1 \cup M_2 \in \mathcal{U}$ and $M_2 \notin \mathcal{U}$ implies $M_1 \in \mathcal{U}$ hence $a_1 \in \mathfrak{g}_{\mathcal{U}}$. It is also clear that $\mathfrak{g}_{\mathcal{U}} \in D(f)$ if and only if $D(f) \in \mathcal{U}$. By Lemma 1.1, $\mathfrak{g}_{\mathcal{U}}$ is $\text{glim } \mathcal{U}$ and all $D(f)$ are glim -closed. $\square$

Lemma 1.1. *Let $\mathcal{B}$ be a topology base of the topological space X .*

- a) *Let $\mathcal{U}$ be an ultrafilter on X and $g \in X$ such that for all $\Omega \in \mathcal{B}$, $g \in \Omega \Leftrightarrow \Omega \in \mathcal{U}$. Then g is a generic limit of $\mathcal{U}$.*
- b) *If X is T_0 and for every ultrafilter $\mathcal{U}$ on X , there is $g \in X$ as in a), then X is spectral and the elements of $\mathcal{B}$ are quasi-compact.*

Proof. Let $U \in \mathcal{N}_g$. Then there is $\Omega \in \mathcal{B}$ such that $g \in \Omega \subseteq U$. By our assumption, $\Omega \in \mathcal{U}$ hence $U \in \mathcal{U}$ thus g is a limit of $\mathcal{U}$. Let $x \notin \overline{\{g\}}$ then there is $U \in \mathcal{N}_x$ such that $g \notin U$. As $\mathcal{B}$ is a topology base there is $\Omega \in \mathcal{B}$ such that $x \in \Omega \subseteq U$. Then $g \notin \Omega$, hence $\Omega \notin \mathcal{U}$. Since Ω is a neighbourhood of x , this shows $x \notin \text{Lim } \mathcal{U}$. Thus g is a generic limit. This proves a).

For b), it follows from our assumption and a) that X has unique generic limits of ultrafilters and the elements of $\mathcal{B}$ are closed under such generic limits. The claim now follows from Proposition 1.30. $\square$

Remark 1.16. *If $\mathcal{U}$ is an ultrafilter on $\text{Spec } A$ and*

$$\mathfrak{g}_{\mathcal{U}} = \{a \in A \mid \{\mathfrak{p} \in \text{Spec } A \mid a \in \mathfrak{p}\} \in \mathcal{U}\}.$$

Then the lemma applies with $\mathcal{B} = \{D(a) \mid a \in A\}$.

Corollary 1.2 (Of Lemma 1.1 b), and the proof of Theorem 1.2). *The subset $D(a) \subseteq \text{Spec } A, a \in A$ are quasi-compact.*

Proof. From Remark 1.16 and Lemma 1.1. $\square$

Definition 1.29. *Let X be a topological space. A subset U which is open and closed is called clopen.*

Corollary 1.3 (Of Proposition 1.30). *For a compact (thus Hausdorff) X , the following conditions are equivalent.*

a). X is spectral.

b). The clopen subsets of X form a topology base.

In this case, the open subsets U is quasi-compact if and only if it is clopen and for a spectral space Y , a map $Y \rightarrow X$ is spectral it is continuous.

Proof. Every ultrafilter $\mathcal{U}$ of X admits a unique limit by Proposition 1.25 and Proposition 1.26. Such limits are trivially generic limits. By Corollary 1.1, the open subsets which are invariant under glim are closed.

To show these two are equivalent. If U is closed, it is quasi-compact as it is a closed subset of a quasi-compact space. If U is quasi-compact then it is closed by Proposition 1.15. If $Y \xrightarrow{f} X$ is continuous and $\Omega \in \mathfrak{Q}_C(X)$, then Ω is clopen hence $f^{-1}(\Omega)$ is clopen in Y , hence $f^{-1}(\Omega) \in \mathfrak{Q}_C(Y)$. $\square$

Definition 1.30. A set $\mathcal{A}$ of subsets of X is a boolean algebra if the following conditions hold.

1. $\emptyset \in \mathcal{A}$.
2. $M \in \mathcal{A} \Rightarrow X \setminus M \in \mathcal{A}$.
3. $\mathcal{A}$ is closed under taking finite unions.

Remark 1.17. From the basic axioms of boolean algebras. Clearly we have $X \in \mathcal{A}$ and $M_1, M_2 \in \mathcal{A} \Rightarrow M_1 \cap M_2 \in \mathcal{A}$.

Proposition 1.31. Let X be a spectral space. The following boolean algebras of subsets of X coincide.

1. The boolean algebra generated by $\mathfrak{Q}_C(X)$.
2. $\{M \subseteq X \mid M \text{ is closed under generic limits of ultrafilters}\}$.

Let $\mathcal{K}_X$ be this boolean algebra of subsets of X . The topology on X obtained by taking $\mathcal{K}_X$ as a topology base coincides with the one for which taking the subsets $M \subseteq X$ which are closed under glim of ultrafilters as closed subsets.

Finally let X_{con} be X equipped with this topology. Then X_{con} is a spectral hausdorff space (i.e. X_{con} as in Corollary 1.2), and $X_{\text{con}} \xrightarrow{\text{id}_X} X$ is a spectral map.

For every ultrafilter $\mathcal{U}$ of X , $\lim_{X_{\text{con}}} \mathcal{U} = \text{glim}_X \mathcal{U}$.

Proof: (Caution : This is not required for the exam). Let $\mathcal{K}_1$ be the Boolean algebra generated by $\mathfrak{Q}_C(X)$ and

$$\mathcal{K}_2 := \{M \mid M \text{ and } X \setminus M \text{ closed under } \text{glim}\}.$$

By Proposition 1.28, $\mathcal{K}_2$ is a Boolean algebra. By Proposition 1.30 b), and Corollary 1.1, $\mathfrak{Q}_C(X) \subseteq \mathcal{K}_2$, hence we have an inclusion $\mathcal{K}_1 \subseteq \mathcal{K}_2$. For the time being, $\mathcal{K}_x := \mathcal{K}_1$. Let X_1 be the topological space with underlying space X and

the topology generated by $\mathcal{K}_x$. Let X_2 be the topological space again with the same underlying set such that for a subset $A \subseteq X$ is closed if and only if it is closed under glim of ultrafilters. By Proposition 1.28, X_2 is indeed a topological space. Since $\mathcal{K}_x \subseteq \mathcal{K}_2$, X_2 is finer or equal to X_1 .

The space X_1 is Hausdorff. If $x \neq y$ then without loss of generality exchanging x and y if necessary, there is $U \in \mathcal{O}_X$ such that $x \in U$ and $y \notin U$. As $\mathfrak{Q}_C(X)$ is a topology base, hence without loss of $U \in \mathfrak{Q}_C(X)$. Then $U \in \mathcal{K}_X$ and $X \setminus U \in \mathcal{K}_X$ and they are distinguished neighbourhoods of x and y in X_1 . Moreover, X_2 is quasi-compact by Proposition 1.25. Thus it suffices to consider an ultrafilter $\mathcal{U}$ on X and to show that it has a limit in X_2 . Let $g = \text{glim}_X \mathcal{U}$. We claim that $g \in \text{Lim}_{X_2} \mathcal{U}$. For this, let $U \in \mathcal{O}_{X_2}$ such that $g \in U$, we must show $U \in \mathcal{U}$. Otherwise $X \setminus U \in \mathcal{U}$. But by the definition of X_2 , $X \setminus U$ is closed under glim_X of ultrafilters. Thus $g \in X \setminus U$ which is a contradiction.

By Proposition 1.15, the continuous map $X_2 \xrightarrow{\text{id}_X} X$ is a homeomorphism. Thus first two parts are shown.

Let $X_{\text{con}} = X_1 = X_2$. Let $\Omega \in \mathcal{K}_2$. By Corollary 1.1, Ω is clopen in X_{con} . Since $\mathcal{K}_1$ is a topology base, Ω is a union of elements of $\mathcal{K}_1$ and since $\Omega \in \mathfrak{Q}_C(X_{\text{con}})$, the union can be chosen to be finite. Then $\Omega \in \mathcal{K}_1$ as $\mathcal{K}_1$ is a Boolean algebra of subsets.

From now on, we will regard $\mathcal{K}_x = \mathcal{K}_1 = \mathcal{K}_2$. Since the elements of $\mathcal{K}_x$ are clopen and $\mathcal{K}_x$ is a topology base for X_{con} , and X_{con} compact. By Corollary 1.3, X_{con} is spectral. By Proposition 1.30, $X_{\text{con}} \xrightarrow{\text{id}_X} X$ is spectral. $\square$

Definition 1.31. *The elements of $\mathcal{K}_X$ are called constructible subsets and the topology of X_{con} the constructible topology.*

Corollary 1.4. *A subset of X is an element of $\mathcal{K}_X$ if and only if it is clopen in X_{con} .*

Corollary 1.5 (Of Proposition 1.30). *A map $X \xrightarrow{f} Y$ between spectral spaces is continuous if and only if spectral is continuous and $X_{\text{con}} \xrightarrow{f} Y_{\text{con}}$ is continuous.*

Corollary 1.6. *For a spectral space X , $\mathfrak{Q}_C(X) = \mathcal{O}_X \cap \mathcal{K}_X$.*

Definition 1.32. *Let X be a spectral. We say that $x \in X$ is a specialization of $y \in X$ and conversely y is a generalization of x if $x \in \overline{\{y\}}$. We denote such relation by $x \preceq y$.*

Proposition 1.32.

1. *A subset $Z \subseteq X$ is closed in X if and only if it is closed in X_{con} and closed under specialization. That is $x \preceq y, y \in Z \Rightarrow x \in Z$.*
2. *A subset $U \subseteq X$ is open in X if and only if it is open in X_{con} and closed under generalization.*
3. *Let Z be closed in X_{con} and $\overline{Z}$ be its closure in X . Then $\overline{Z}$ is the set of specialization of elements of Z .*

Proof. If $z \in Z$ and $x \preceq z$, then $x \in \overline{\{z\}} \subseteq \overline{Z}$, there are ???. If $x \in \overline{Z}$ by Proposition 1.23, there is an ultrafilter $\mathcal{U}$ on X such that $Z \in \mathcal{U}$ and $x \in \text{Lim} \mathcal{U}$. Since Z is closed in X_{con} , $g = \text{glim} \mathcal{U} \in Z$. But $x \preceq g$, proving the third statement. The first two are equivalent and the first follows from the third. $\square$

Corollary 1.7. *For a spectral space X , the following statements are equivalent.*

1. X is T_0 .
2. $X = X_{\text{con}}$ as topological space.
3. Every quasi-compact subset of X is closed.
4. X is Hausdorff.
5. X is T_4 .

Corollary 1.8. *For a map $X \xrightarrow{f} Y$ between spectral spaces, the following statements are equivalent.*

1. $\text{glim}_Y f_* \mathcal{U} = f(\text{glim}_X \mathcal{U})$ for all ultrafilters $\mathcal{U}$ on X , and $x \preceq_X \xi$ implies $f(x) \preceq_Y f(\xi)$.
2. $X_{\text{con}} \xrightarrow{f} Y_{\text{con}}$ and $x \preceq_X \xi$ implies $f(x) \preceq_Y f(\xi)$.
3. f is spectral.

Corollary 1.9. *For a subset $Z \subseteq X$, the following statements are equivalent.*

- a). Z is spectral with induced topology and $Z \rightarrow X$ is spectral.
- b). X is closed in X_{con} .

In this case, we denote Z_{con} to be the topology on Z induced from topology X_{con}

1.4 The Valuation Spectrum of a Rings

Definition 1.33. *An ordered abelian group is an abelian group Γ (under multiplication) with a linear ordering $\leq$ such that*

$$\gamma_1 \leq \gamma_2, g_1 \leq g_2 \Rightarrow \gamma_1 g_1 \leq \gamma_2 g_2.$$

Example 1.7. $\{1\}$ and $(0, \infty)_{\mathbb{R}}^\times$ are ordered abelian groups.

Remark 1.18. *The inequality,*

$$\gamma_1 \leq \gamma_2, g_1 \leq g_2 \Rightarrow \gamma_1 g_1 \leq \gamma_2 g_2.$$

is strict if one of the inequalities in the left hand side is strict. In particular, $g > 1$ implies $g^k > 1$ and $g < 1$ implies $g^k < 1$ for all $k \in \mathbb{N}$. Thus Γ has no torsion.

Definition 1.34. A (general) valuation of a ring R is a map $R \xrightarrow{\gamma} \Gamma \cup \{0\}$ where Γ is an ordered abelian group.

Definition 1.35. A valuation of a ring R is a map $R \xrightarrow{\gamma} \Gamma \cup \{0\}$ where Γ is an ordered abelian group with multiplication such that the following hold.

$$i). \forall x \in \Gamma \cup \{0\}, 0 \cdot x = x \cdot 0 = 0, 0 \leq x.$$

$$ii). \gamma(xy) = \gamma(x)\gamma(y)$$

$$iii). \gamma(1) = 1$$

$$iv). \gamma(0) = 0$$

$$v). \gamma(x + y) \leq \max\{\gamma(x), \gamma(y)\}.$$

Remark 1.19.

$$\gamma(-1) = 1, \quad \gamma(y) < \gamma(x) \Rightarrow \gamma(x + y) = \gamma(x).$$

Definition 1.36. Let Γ_γ be the subgroup of Γ generated by $\{\gamma(r) \mid r \in R\} \setminus \{0\}$. We say that valuations γ and ω are equivalent if there is an order preserving isomorphism $\Gamma_\gamma \xrightarrow{i_\sim} \Gamma_\omega$ such that

$$\omega(r) = i(\gamma(r)),$$

where we put $i(0) = 0$.

Proposition 1.33. The valuation of r and w are equivalent if and only if

$$\{(r, s) \in R^2 \mid \gamma(r) \leq \gamma(s)\} = \{(r, s) \in R^2 \mid w(r) \leq w(s)\}.$$

Definition 1.37.

$$\text{supp } \gamma = \{r \in R \mid \gamma(r) = 0\}.$$

Remark 1.20. $\text{supp } \gamma$ is a prime ideal. Indeed, $\gamma(r) = 0$ then $\gamma(r, s) = \gamma(r)\gamma(s) = 0$ with $s \in R$, if $r, s \in \text{supp } \gamma$ then $\gamma(r + s) \leq \max\{\gamma(r), \gamma(s)\} = 0$. Clearly $1 \notin \text{supp } \gamma$. If $\gamma(rs) = 0$, then $\gamma(r)\gamma(s) = 0$. Thus $\gamma(r) = 0$ or $\gamma(s) = 0$.

Remark 1.21. The support only depends on the equivalence classes.

Example 1.8. If $\mathfrak{p}$ is a prime ideal and

$$\gamma(r) = \begin{cases} 1, & r \notin \mathfrak{p}, \\ 0, & \text{quadr} \in \mathfrak{p}. \end{cases}$$

the trivial valuation belonging to $\mathfrak{p}$.

Definition 1.38. A valuation is trivial if it is equivalent to this kind of valuation. That is $\Gamma_\gamma = \{1\}$.

Example 1.9. If $R = K$ is a field, then $\gamma(s) = 1$ where $\gamma^k = 1$ for some $k \in \mathbb{N}$ as Γ has no torsion. Then for a finite field only admits trivial valuations.

Example 1.10. The p -adic valuation of $\mathbb{Q}$ or $\mathbb{Q}_p$.

Definition 1.39. Let $\text{Spv}(R)$ be the set of equivalent classes of valuation of R . For $x \in \text{Spv}(R)$, we choose a representative $|\cdot|_x$. Let Γ_x be the group generated by those $|r|_x$ which are non-zero. For $m \in \mathbb{N}$ and $(r_i)_{i=0}^m \in \mathbb{R}^{m+1}$, let

$$\mathcal{R}(r_0|r_1, \dots, r_m) = \mathcal{R}_{\text{Spv}(R)}(r_0|r_1, \dots, r_m) = \{x \in \text{Spv}(R) | 0 \neq |f_0|_x \geq |f_i|_x, 1 \leq i \leq m\}.$$

These are called rational open subsets. We equip $\text{Spv}(R)$ with the topology for which the set of rational open subsets is a topology base.

If $A \xrightarrow{\alpha} B$ is a ring homomorphism then this induces

$$\text{Spv}(B) \xrightarrow{\text{Spv}(\alpha)} \text{Spv}(A)$$

sends $x \in \text{Spv}(B)$ to the equivalence class of the valuation,

$$a \mapsto |\alpha(a)|_x$$

of A . We set,

$$\gamma(a) = \text{Spv}(A) \setminus \mathcal{R}(a).$$

Remark 1.22. The topology there is the coordinated? for which

$$\text{Spv} \xrightarrow{x \mapsto \text{supp}(x)} \text{Spec}$$

is continuous and all

$$\{x \in \text{Spv}(A) \setminus \mathcal{R}(a) \mid |\alpha|_x \leq |a|_x\}$$

for $(a, \alpha) \in A^2$ are open.

Remark 1.23. The set of rational open subset is closed under finite intersections in $\text{Spv}(R)$. Indeed,

$$\begin{aligned} \mathcal{R}(1) &= \text{Spv}(R), \\ \mathcal{R}(r_0|r_1, \dots, r_m) \cap \mathcal{R}(s_0|s_1, \dots, s_n) &= \mathcal{R}(r_0 s_0 | r_i s_i, 0 \leq i \leq n, 0 \leq j \leq n). \end{aligned}$$

If you assume $m \leq 1$ or even $m = 1$, you only get a topology subbase.

One way of constructing generic limits of ultrafilters in $\text{Spv}(R)$ would be to use Proposition 1.33 and

$$\{(r, s) \in R^2 \mid |r|_g \leq |s|_g\} := \{(r, s) \in R^2 \mid \{x \in \text{Spv}(R) \mid |r|_x \leq |s|_x\} \in \mathcal{U}\}.$$

This would work and was shown by Huber.

Definition 1.40. Let $\mathcal{U}$ be an ultrafilter on Λ and $(X_\lambda)_{\lambda \in \Lambda}$ be a family of sets indexed by Λ . We call the elements $(x_\lambda)_{\lambda \in \Lambda}$ and $(\xi_\lambda)_\Lambda$ of $\prod_{\lambda \in \Lambda} X - \lambda$ $\mathcal{U}$ -equivalent if

$$\{\lambda \mid x_\lambda = \xi_\lambda\} \in \mathcal{U}.$$

The ultraproduct $\prod_{\mathcal{U}} X_\lambda$ is the set of $\mathcal{U}$ -equivalence classes of elements of $\prod_{\lambda \in \Lambda} X_\lambda$. If X_λ -s have group structures, $(x\xi) \in \prod_{\mathcal{U}} X_\lambda$ is given as the $\mathcal{U}$ -equivalence classes of $(x_\lambda \xi_\lambda)_\Lambda$. If X_λ -s are equipped with linear ordering, put

$$(x_\lambda)_{\lambda \in \Lambda} / \sim_{\mathcal{U}} \leq (\xi_\lambda)_{\lambda \in \Lambda} / \sim_{\mathcal{U}}$$

if $\{\lambda \in \Lambda \mid x_\lambda \leq \xi_\lambda\} \in \mathcal{U}$.

Proposition 1.34. Suppose $X_\lambda \neq \emptyset$ for all $\lambda \in \Lambda$.

1. If $\mathcal{U} = \{\Theta \subseteq \Lambda \mid \lambda_0 \in \Theta\}$ is the principal ultrafilter belonging to λ_0 , then $\prod_{\mathcal{U}} X_\lambda = X_{\lambda_0}$.
2. If F is a finite set and all $\lambda \in \Lambda, X_\lambda = F$. Then $\prod_{\mathcal{U}} F = F$ as $\Lambda = \bigcup_{f \in F} \{\lambda \mid x_\lambda = f\}$ (i.e. disjoint finite union hence there is $??$ are $f \in F$ such that $\{\lambda \mid x_\lambda = f\} \in \mathcal{U}$ where $(x)_\lambda = \prod_\Lambda F$)
3. If $X_\lambda = X'_\lambda \cup X''_\lambda$ (disjoint unions, $\forall \lambda \in \Lambda, X'_\lambda, X''_\lambda \neq \emptyset$) Then $\prod_{\mathcal{U}} (X'_\lambda \cup X''_\lambda) = (\prod_{\mathcal{U}} X'_\lambda) \cup (\prod_{\mathcal{U}} X''_\lambda)$ which is also a disjoint union as $\Lambda = \{\lambda \in \Lambda \mid x_\lambda \in X'_\lambda\} \cup \{\lambda \in \Lambda \mid x_\lambda \in X''_\lambda\}$.
4. Then $\prod_{\mathcal{U}} (\Gamma_\lambda \cup \{0\}) = (\prod_{\mathcal{U}} \Gamma_\lambda) \cup \{0\}$ and $\prod_{\mathcal{U}} \Gamma_\lambda$ is an ordered abelian group where all Γ_λ are ordered abelian groups.

Remark 1.24. When some of the X_λ may be empty, it is better to define $\prod_{\mathcal{U}} X_\lambda$ as the set of equivalent classes of pairs (M, ξ) where $M \in \mathcal{U}$ is a subset and ξ a function defined on M such that $\xi(\lambda) \in X_\lambda$ where $\lambda \in M$. Two pairs $(M, \xi), (N, \nu)$ are equivalent if

$$\{\lambda \in M \cap N \mid \xi(\lambda) = \nu(\lambda)\} \in \mathcal{U}.$$

With this, we define,

$$\prod_{\mathcal{U}} \left(\bigcup_{i=1}^m X_\lambda^{(i)} \right) \cong \bigcup_{i=1}^m \prod_{\mathcal{U}} X_\lambda^{(i)}.$$

Remark 1.25. $\prod_{\mathcal{U}}(\text{ordered abelian group})$ is again an ordered abelian group and

$$\prod_{\mathcal{U}} (\Gamma_\lambda \cup \{0\}) \cong \left(\prod_{\mathcal{U}} \Gamma_\lambda \right) \cdot \{0\}.$$

Proposition 1.35. For every ring R , $\text{Spv}(R)$ is a spectral space and a subset of the form $\mathcal{R}(f_0 \mid f_1, \dots, f_m)$ with $m \in \mathbb{N}$ and $g(f_i)_{i=0}^m \in R^{m+1}$ is quasi-compact.

The generic limit of the ultrafilter $\mathcal{U}$ on $\text{Spv } R$ is given by the equivalence class of the valuation,

$$R \xrightarrow{r \mapsto (\text{the } \mathcal{U}\text{-equivalence class of } (|r|_\gamma)_{\gamma \in \text{Spv } R})} \left(\prod_{\gamma \in \text{Spv } R_{\mathcal{U}}} \Gamma_\gamma \right) \cup \{0\} = \prod_{\mathcal{U}} (\Gamma_\gamma \cup \{0\}). \quad (1)$$

Moreover, the map,

$$\text{Spv}(R) \xrightarrow{\gamma \mapsto \text{supp } |\cdot|_\gamma} \text{Spec } R$$

is spectral and for every ring homomorphism $R \xrightarrow{\rho} S$, the map

$$\text{Spv}(S) \xrightarrow{\text{Spv}(\rho)} \text{Spv}(R)$$

is also spectral.

Proof. It is obvious from the definition and the properties of ultrafilters that (1) is indeed a valuation. To show that it defines a generic limit of $\mathcal{U}$ and that the rational open subsets are quasi-compact, we use Lemma 1.1. Set

$$\mathcal{B} = \{\mathcal{R}(f_0 | f_1, \dots, f_m) \mid m \in \mathbb{N}, (f_i)_{i=0}^m \in R^{m+1}\}.$$

if g ??? the equivalence class of (1),

$$[g \in \mathcal{R}(f_0 | f_1, \dots, f_m)] \Leftrightarrow |f_0|_g \neq 0 \text{ and all } |f_1|_g \leq |f_0|_g,$$

$$\stackrel{\text{By (1)}}{\Leftrightarrow} (|f_0|_x)_{x \in X} / \sim_{\mathcal{U}} \neq 0 \text{ and } (|f_i|_x)_{x \in X} / \sim_{\mathcal{U}} \leq (|f_0|_x)_{x \in X} / \sim_{\mathcal{U}},$$

$$\stackrel{\substack{\text{Definition of } \prod_{\mathcal{U}} \\ \text{and the valuation on it}}}{\Leftrightarrow} \{x \in X \mid |f_0|_x \neq 0\} \in \mathcal{U} \text{ and all } \{x \in X \mid |f_i(x)| \leq |f_0(x)|\} \in \mathcal{U},$$

$$\stackrel{\cap_{i=0}^m M_i \in \mathcal{U} \Leftrightarrow \forall M_i \in \mathcal{U}}{\Leftrightarrow} \{x \in X \mid |f_0|_x \neq 0\} \cap \bigcap_{i=1}^m \{x \in X \mid |f_i(x)| \leq |f_0(x)|\} \in \mathcal{U},$$

$$\stackrel{\text{Definition of rational open}}{\Leftrightarrow} \mathcal{R}(f_0 | f_1, \dots, f_m) \in \mathcal{U}.$$

Lemma 1.1 can indeed be applied. That the induced map $\text{Spv}(R) \xrightarrow{\text{supp}} \text{Spec}(R)$ is spectral follows from

$$\text{supp}(D(f)) = \mathcal{R}(f|).$$

Proposition ???. That $\text{Spv}(\rho)$ is spectral follows immediately from

$$\text{Spv}(\rho)^{-1} \mathcal{R}(f_0 | f_1, \dots, f_m) = \mathcal{R}(\rho(f_0) | \rho(f_1), \dots, \rho(f_m)).$$

□

Remark 1.26.

Proposition 1.36. *For every ideal $I \subseteq R$, if $R \xrightarrow{\pi} R/I$ denotes the canonical projection, then the induced map $\mathrm{Spv}(R/I) \xrightarrow{\pi} \mathrm{Spv}(R)$ is a homeomorphism. Explicitly, we have,*

$$\mathrm{Spv}(R/I) = V_{\mathrm{Spv}(R)}(I) := \{x \in \mathrm{Spv}(R) \mid \mathrm{supp} x \supseteq I\} = \mathrm{Spv}(R) \setminus \bigcup_{f \in I} \mathcal{R}(f). \quad (1.3)$$

If I is finitely generated (i.e. $I = \langle g_1, \dots, g_m \rangle_R$), then the right hand side of Equation (1.3) can be written as,

$$\mathrm{Spv}(R/I) = \bigcup_{i=1}^m \mathcal{R}(g_i)$$

which is the embedded subset. Alternatively, we have,

$$V_{\mathrm{Spv}(R)} = \{x \in \mathrm{Spv}(R) \mid \forall f \in I, |f|_x = 0\}.$$

Furthermore, we can identify this with a homeomorphism,

$$\mathrm{Spv}(R/I) \xrightarrow[\simeq]{\mathrm{Spv}(\pi)} V(I).$$

Let S be a multiplicative subset of S and $\iota : R \rightarrow R_S$ be a canonical inclusion then,

$$\mathrm{Spv}(R_S) = \xrightarrow[\mathrm{Spv}(\iota)]{\simeq} \bigcap_{s \in S} (\mathrm{Spv}(R) \setminus V(s)) = \bigcap_{s \in S} \mathcal{R}(s). \quad (1.4)$$

Proposition 1.37. *Let s be a specialization of $\eta \in \mathrm{Spv}(R)$, then*

$$\mathrm{supp} \eta \subseteq \mathrm{supp} s. \quad (1.5)$$

Proof.

$$r \notin \mathrm{supp} s \iff s \in \mathcal{R}(r) \Rightarrow \eta \in \mathcal{R}(r) \iff r \notin \mathrm{supp} \eta. \quad (1.6)$$

□

Definition 1.41. *Let $\gamma \in \Gamma \cup \{0\}$ and $R \xrightarrow{v} \Gamma \cup \{0\}$ be a valuation. We then define,*

$$K_{v \leq \gamma} := \{r \in R \mid v(r) \leq \gamma\}.$$

For an equivalence class $x \in \mathrm{Spv}(R)$ where $v \in x$, we may denote,

$$K_{x \leq \gamma} = K_{v \leq \gamma}.$$

For $\rho \in R$, we denote,

$$K_{v \leq \rho} := K_{v \leq v(\rho)}, K_{x \leq \rho} := K_{x \leq |\rho|_x}.$$

Similarly,

$$\begin{aligned} K_{v < \gamma} &= K_{x < \gamma} := \{r \in R \mid v(r) < \gamma\} \\ K_{v < \rho} &= K_{x < \rho} := K_{v < v(\rho)}. \end{aligned}$$

Proposition 1.38. $K_{v \leq 1}$ is an integrally closed subring of R , and $K_{v < 1}$ is a prime ideal in $K_{v \leq 1}$.

Proof. These being a subring and an ideal can be checked easily. Let $r \notin K_{v \leq 1}$, then if m is a positive integer and $(p_i)_{i=1, \dots, m} \in K_{v \leq 1}^m$, we have,

$$\begin{aligned} v(r^m) &= v(r)^m, \\ &> v(r)^{m-1} \leq \max_{1 \leq i \leq m-1} v(p_i) v(r)^i, \\ &= \max_{1 \leq i \leq m-1} v(p_i r^i), \\ &\leq v \left(\sum_{i=1}^{m-1} p_i r^i \right). \end{aligned}$$

Hence,

$$r^m \neq \sum_{i=1}^{m-1} p_i r^i. \quad (1.7)$$

Thus r is not integral over $K_{v \leq 1}$, in particular, $K_{v \leq 1}$ is integrally closed.

For $K_{v < 1}$ being prime, obviously $1 \notin K_{v < 1}$. Let $a, b \in K_{v \leq 1} \setminus K_{v < 1}$. Then $v(a) = v(b) = 1$. Therefore $v(ab) = 1$. Showing $K_{v < 1}$ is a prime ideal. $\square$

Proposition 1.39. Let s be a specialization of $\eta \in \text{Spv}(R)$. We have,

$$a \in R \setminus \text{supp } s \Rightarrow K_{\eta < a} \subseteq K_{s < a}.$$

Applying this to $a = 1$, (recall that $\text{supp } s$ is a prime ideal), we get,

$$K_{\eta < 1} \subseteq K_{s < 1}.$$

Proof. By the assumption and Proposition 1.37, we know that $|a|_\eta \neq 0$. Therefore,

$$b \notin K_{\eta < a} \iff \eta \notin \mathcal{R}(b|a) \Rightarrow s \notin \mathcal{R}(b|a) \iff b \notin K_{s < a}.$$

$\square$

Proposition 1.40. Let s be a specilization of $\eta \in \text{Spv}(R)$.

$$a \in R \setminus \text{supp}(s) \Rightarrow K_{s \leq a} \subseteq K_{\eta \leq a}.$$

In particular for $a = 1$, we get,

$$K_{s \leq 1} \subseteq K_{\eta \leq 1}.$$

Proof. By Proposition 1.37, $|a|_\eta \neq 0$. Hence,

$$b \in K_{s \leq a} \iff s \in \mathcal{R}(a|b) \Rightarrow \eta \in \mathcal{R}(a|b) \iff b \in K_{\eta \leq a}.$$

$\square$

Definition 1.42. Let X be a linearly ordered set. For $a, b \in X$ with $a \leq b$, we define the line segment between a and b to be

$$|x, y|_X := \{x \in X \mid a \leq x \leq b\}.$$

A subset $C \subseteq X$ is convex if for any two elements $c, d \in C$, $c \leq d$, we have $|c, d|_X \subseteq C$.

Definition 1.43. A convex subgroup Θ is a subgroup of a linearly ordered group Γ which is convex.

Proposition 1.41. A subgroup Θ of a linearly ordered group Γ is convex if and only if for any $\theta \in \Theta$ with $\theta < 1$, $|\theta, \theta^{-1}|_\Gamma \subseteq \Theta$.

Recall that for a valuation $v : R \rightarrow \Gamma \cup \{0\}$, Γ_v is the subgroup of Γ generated by $v(R) \setminus \{0\}$.

Definition 1.44. $c\Gamma_v$ is the convex subgroup of Γ_v generated by $\{v(r) \mid r \in R, v(r) \geq 1\}$.

Remark 1.27. Equivalently, we can define $c\Gamma_v$ to be a subgroup generated by $\{v(r) \mid r \in R, v(r) > 1\}$.

Proposition 1.42.

$$c\Gamma_v = \{\gamma \in \Gamma_v \mid \exists r \in R \text{ s.t. } \min\{\gamma, \gamma^{-1}\}v(r) \geq 1\}$$

Proof. Let $c\Gamma'_v = \{\gamma \in \Gamma_v \mid \exists r \in R \text{ s.t. } \min\{\gamma, \gamma^{-1}\}v(r) \geq 1\}$. We will show that $c\Gamma_v = c\Gamma'_v$. If $r \in R$ is as in the construction of $c\Gamma'_v$, then $v(a) \geq 1$ thus $v(a) \in c\Gamma_v$. If $\gamma \in c\Gamma'_v$, then

$$v(a)^{-1} \leq \gamma \leq v(a).$$

Hence $\gamma \in c\Gamma_v$. In particular $c\Gamma'_v \subseteq c\Gamma_v$. On the other hand $c\Gamma'_v$ is a convex subgroup of Γ_v containing all $\gamma(r)$ such that $\gamma(r) \geq 1$. Thus by the minimality of $c\Gamma_v$, we have shown the statement. $\square$

Remark 1.28. If H is a convex subgroup of Γ_v , then $c\Gamma_v \subseteq H$.

Remark 1.29. If R is a field then $c\Gamma_v = \Gamma_v$.

Definition 1.45. Let $A \xrightarrow{v} \Gamma$ be a valuation of A and $H \subseteq \Gamma_v$ a convex subgroup containing $c\Gamma_v$. Let

$$v|_H(r) := \begin{cases} v(r) & (v(r) \in c\Gamma_v) \\ 0 & (\text{otherwise}). \end{cases}$$

Proposition 1.43. Under the assumption of Definition 1.42, $v|_H$ is indeed a valuation of A and $v|_H$ is a specialization of v in $\text{Spv}(A)$.

Proof. Let M be the convex hull of $\{0\} \cup H$ in $\Gamma_v \cup \{0\}$. For $m \in M$, set,

$$\varphi(m) = \begin{cases} m, & (m \in H), \\ 0, & (\text{otherwise}). \end{cases}$$

Since $H \supseteq \Gamma_v$, we have $v(a) \in M$ for all $a \in A$. (note that $[0, 1]_{\Gamma_v \cup \{0\}} \supseteq M$ and $v(a) \in c\Gamma_v \subseteq H \subseteq M$ where $v(a) \geq 1$) Since r is ??? and $\varphi(0) = 0$, and $\varphi(1) = 1$, the axiom,

$$w(0) = 0, w(1) = 1, w(a + b) \leq \max\{w(a), w(b)\}$$

for $w = v|_H = \varphi r$ follow from their counterparts for v . Also M is multiplicative ?? and $\varphi(m\mu) = \varphi(m)\varphi(\mu)$. (This is obvious when $\varphi(m)\varphi(\mu) \neq 0$. Otherwise, without loss of generality $\varphi(m) = 0$, hence $m \notin H$. In particular, $m < 1$. Also $hm < 1$ for all $h \in H$ by the ??? of H . Thus, $nm < 1$ for all $n \in M$, in particular $m\mu < 1$. Now $m\mu \in H$ would imply $\mu \in H$ as $m\mu \leq \mu$ and $\mu \in M$ (hence μ is bounded from above by some $h \in H$). But then $m \in H$ a contradiction).

To see w is a specialization of r , observe that

$$\begin{aligned} w \in \mathcal{R}(f_0|f_1, \dots, f_m) &\iff [w(f_0) = 0 \text{ and } w(f_0) \geq w(f_i), i = 1, \dots, m], \\ &\iff [v(f_0) \in H \text{ and } v(f_0) \geq w(f_i), i = 1, \dots, m], \\ &\iff [v(f_0) \in H \text{ and } v(f_0) \geq v(f_i), i = 1, \dots, m]. \end{aligned}$$

Note that $v(f_i) > v(f_0) \in H$ would imply $v(f_i) \in H$ by the conexity of H , as $v(f_i) \in H$, hence $v(f_i)$ bounded from above by some elements of H . Hence $w(f_i) = v(f_i) > v(f_0) = w(f_0)$ if $v(f_i) > v(f_0)$. Furthermore,

$$\Rightarrow [v(f_0) \neq 0 \text{ and } v(f_i) \leq v(f_0), i = 1, \dots, m] \iff (v/\sim) \in \mathcal{R}(f_0|f_1, \dots, f_m).$$

□

Remark 1.30. Motivated by the above notion, we say $s \in \text{Spv}(R)$ is a horizontal specialization of $\eta \in \text{Spv}(R)$ if there is $v \in \eta$ and a convex subgroup $c\Gamma_v \subseteq H \subseteq \Gamma_v$ such that s is equivalent to $v|_H$.

Corollary 1.10. If x is a closed point of $\text{Spv}(A)$, then $c\Gamma_x = \Gamma_x$.

Proof. If this is not the case, $x|_{c\Gamma_x}$ is a proper specialization of x . □

Proposition 1.44. If $s \in \text{Spv}(A)$ is a horizontal specialization of $\eta \in \text{Spv}(A)$ and $\Omega \in \mathcal{R}(f_0|f_1, \dots, f_m)$ then $s \in \Omega \iff (\eta \in \Omega \text{ and } |f_0|_s \neq 0)$.

Proof. Indeed, the construction of the end of the proof of Proposition 1.43 imply

$$(w/\sim) \in \Omega \iff [(v/\sim) \in \Omega \text{ and } w(f_0) \neq 0].$$

□

Definition 1.46. Let $A \xrightarrow{v} \Gamma \cup \{0\}$ be a valuation and a convex subgroup $H \subseteq \Gamma$ or $H \subseteq \Gamma_v$. The vertical specialization of v with respect to H is

$$(v/H)(a) = \begin{cases} v(a)H \text{ (i.e. a coset represented by } v(a)), & (v(a) \neq 0), \\ 0, & (\text{otherwise}). \end{cases}$$

Remark 1.31. s is a vertical specilization of $\eta \in \text{Spv}(A)$ if and only if s is a specialization of η and $\text{supp } s = \text{supp } \eta$.

Remark 1.32. In Huber's literature, he refers horizontal specialization as primary specialization and the vertical specializations as the secondary specializations.

Remark 1.33. Note that $v|_H$ is a specialization of v while v is a specialization of v/H .

1.5 Valuation Rings

Definition 1.47. Let K be a field. A subring $R \subseteq K$ is a valuation ring of K if for all $x \in K^\times$, $x \in R$ or $x^{-1} \in R$.

Example 1.11. Every discrete valuation rings is a valuation ring of its field of fraction. More concretely, $\mathbb{Z}_p$ is a valuation ring of $\mathbb{Q}_p$.

Example 1.12. If S is a valuation ring of L and $K \subseteq L$ is a subfield then $S \cap K$ is a valuation ring of K .

Example 1.13. If $\mathcal{L}$ is a set of valuation rings which is linearly ordered by $\subseteq$. Then $\bigcap_{R \in \mathcal{L}} R$ is a valuation ring of K (if $\mathcal{L} = \emptyset$ then the intersection would be K itself which is trivially a valuation ring) and if $\mathcal{L} \neq \emptyset$ then $\bigcup_{R \in \mathcal{L}} R$ is also a valuation ring of K .

Example 1.14. If R is a valuation ring of K and $R \subseteq S \subseteq K$ then S is a valuation ring of K .

Proposition 1.45. Let R be a valuation ring of K . We have the following.

- a). The field of fractions of R is K .
- b). For $a, b \in R$, either $a \mid b$ or $b \mid a$.
- c). R is a Bézout domain. That is every finitely generated ideals are principal.
- d). R is integrally closed in K .
- e). R is Noetherian if and oly if R is a discrete valuation ring if and only if the unique maximal ideal $\mathfrak{m}_R$ is principal and this is equivalent to R being a principal ideal domain.
- f). The set of ideals of R is linearly ordered by the inclusion.

- g). $\text{Spv}(R)$ is linearly ordered by specialization.
- h). An ideal $I \subseteq R$ is prime if and only if $I = \sqrt{I}$ and $1 \notin I$.
- i). We have an order reversing bijection such that

$$\text{Spec}(R) \xrightleftharpoons[R \cap \mathfrak{m}_S \leftarrow S]{\mathfrak{p} \mapsto R_{\mathfrak{p}}} \{S \mid R \subseteq S \subseteq K \text{ and } S \text{ is a subring of } K\}$$

- j). The set of S such that $R \subseteq S \subseteq K$ and S is a subring of K is linearly ordered by inclusion.

Proposition 1.46. If R and S are valuation ring of K then $R \subseteq S$ is equivalent to $\mathfrak{m}_S \subseteq \mathfrak{m}_R$.

Proof.

$$\mathfrak{m}_R = \{0\} \cup \{x \in K^\times \mid x \notin R\}.$$

□

Proposition 1.47. We have following bijections,

$$\text{Spv}(K) \xrightleftharpoons[v_R / \sim \leftarrow R]{x \mapsto \{k \in K \mid |k|_x \leq 1\}} \{\text{valuation rings } R \text{ of } K\}$$

where

$$v_R(x) = \begin{cases} x \cdot R^\times, & (x \neq 0), \\ 0, & (x = 0). \end{cases}, \Gamma_R = K^\times / R^\times.$$

The ordering on Γ_R is given as follows,

$$xR^\times \leq yR^\times \iff x \in yR.$$

Remark 1.34. Under the bijection in Proposition 1.47, if the embedding of K into a field extension L is denoted i , $\text{Spv}(i)(x_R) = x_{K \cap R}$ where R is a valuation ring of L , $x_R \in \text{Spv}(L)$. Moreover, $x_R \in \mathcal{R}(f_0|f_1, \dots, f_m)$ if and only if $f_0 = 0$ and $f_i \in f_0 \mathcal{R}$.

Proposition 1.48. x_R be a specialization of x_S if and only if $R \subseteq S$.

Proof.

$\Rightarrow$: follows by the preserving f and

$$\begin{aligned} x_R \in \mathcal{R}(f_0|f_1, \dots, f_m) &\iff [f_0 \neq 0 \text{ and } \forall f \in f_0 \mathcal{R}] \\ &\Rightarrow [f_0 \neq 0 \text{ and } \forall f_i \in f_0 S], \\ &\iff [x_S \in \mathcal{R}(f_0|f_1, \dots, f_m)]. \end{aligned}$$

If x_R is a specialization of x_S , then $R = K_{x_R \leq 1} \subseteq K_{x_S \leq 1} = S$ by Proposition 1.40. □

Lemma 1.2. *Let B be an A -algebra and $M \subseteq B$ be a finitely generated A -submodule of B such that $bM = 0$ for any $b \in B$ implies $aM = 0$. If $b \in B$ is such that $bM \subseteq M$ then b is integral over A .*

Proof. Let $(m_i)_{i=1}^l$ be generators of M as an A -module.

$$bm_i = \sum_{j=1}^l \alpha_{ij} m_j.$$

Set $p \in A[T]$ be the characteristic polynomial of (α_{ij}) . Then $p(b) = 0$ by Cauchy-Hamilton. $\square$

Proposition 1.49. *Let A be a subring of the field K . Then the integral closure A in K is equal to the intersection of the valuation rings of K containing A .*

Proof. Let A' be the intersection of all valuation rings containing A . $A' \supseteq A$ is clear as all valuation rings are integrally closed. Thus it is sufficient to consider an element $x \in K$ which is not integral over A and show there is a valuation ring R of K which doesn't contain x . To this end, set

$$\mathcal{M} := \{A \subseteq B \subseteq K \mid B \text{ is an intermediate ring with } x \text{ is not integral over } B\}.$$

Since $A \in \mathcal{M}$, $\mathcal{M} \neq \emptyset$. Let $\mathcal{L} \subseteq \mathcal{M}$ be a non-empty set which is linearly ordered with respect to inclusion. Then $C = \bigcup_{B \in \mathcal{L}} B$ is in $\mathcal{M}$. That is $\mathcal{L}$ is bounded in $\mathcal{M}$. By Zorn's lemma, there is a maximal element R . It remains to show that R is a valuation ring.

Let $k \in K \setminus R$ such that $k^{-1} \notin R$. Then x is integral over $R[x^{\pm}]$, hence there are polynomials $(p_j^{\pm})_{j=0}^{d_{\pm}-1}$ such that

$$x^{d_{\pm}} = \sum_{j=0}^{d_{\pm}-1} x^j p_j^{(\pm)}(k^{\pm 1}).$$

Let $d = \max\{d_+, d_-\}$ and e the maximum of all $\deg p_j^{(\pm)}$ with $0 \leq j < d^{(\pm)}$. Let $N \subseteq K$ be the R -submodule generated by the $x^i k^j$ where $0 \leq i < d$, $0 \leq j < 2e$. Then N is a finitely generated R -module and $xN \subseteq N$. Indeed, consider a generator $y = x^i k^j$ of N . We can assume that $j < e$. If $i < d_+ - 1$ then xy is in the set generators of N . Otherwise,

$$xy = x^{i+1-d_+} \sum_{l=0}^{d_+-1} x^l p_l^{(+)}(k) k^j \in N.$$

Now assume $j \geq e$. If $i < d - 1$, then xy is in the set of generators of N . Otherwise,

$$xy = x^{i+1-d_-} \sum_{l=0}^{d_--1} x^l k^j p_l^{(-)}(k^{-1}) \in N.$$

By Lemma 1.2, x is integral over R . This contradicts that $R \in \mathcal{M}$. $\square$

Remark 1.35. *The book of Kneibuch and Schrecker also has a proof.*

2 G_+ -topological span and their sheaf cohomology

2.1 Grothendieck topologies and G_+ -topological span

Let $\mathcal{C}$ be a category. In the application we are interested in, it will be given by a poset $(P, \preceq)$. Objects of $\mathcal{C}$ are the all of P and a morphism $x \xrightarrow{\mu} y$ exists if and only if $x \preceq y$, in which case μ is unique. In fact $P = \mathcal{O}_X$ (all open subsets of X) or P is some topology base of X , and $\preceq$ is containment $\subseteq$. However, the case where there may be more than one morphism between objects of $\mathcal{C}$ are very important in other applications like étale or crystalline cohomology.

Definition 2.1. A sieve on (or over) an object X of $\mathcal{C}$ or an X -sieve is a class $\mathcal{S}$ of morphisms $U \rightarrow X$ such that $[U \xrightarrow{\nu} X] \in \mathcal{S}$ then $\nu\omega \in \mathcal{S}$ where $V \xrightarrow{\omega} U$ is any morphism in $\mathcal{C}$.

Definition 2.2. If $\mathcal{S}$ is a X -sieve and $Y \xrightarrow{\xi} X$ then $\xi^*\mathcal{S}$ is the Y -sieve of all morphism $U \xrightarrow{\nu} Y$ such that $\xi\nu \in \mathcal{S}$.

Definition 2.3. A Grothendieck topology on $\mathcal{C}$ is a collection $(\Pi_X)_{X \in \text{Ob}(\mathcal{C})}$ of X -sieves (where Π_X being called the class of covering sieves of X) satisfying the following conditions.

(GT-Triviality) The all-sieve $\mathcal{S}$ on X (i.e. the class of all morphisms with its target to X) is in Π_X .

(GT-Transitivity) If $\mathcal{S} \in \Pi_X$ and $Y \xrightarrow{\nu} X$ and any morphism $Y \xrightarrow{\omega} X$ in $\mathcal{C}$, we have $\nu^*\mathcal{S} \in \Pi_Y$.

(GT) If $\mathcal{S} \in \Pi_X$ and any X -sieve $\mathcal{T}$ such that $\nu^*\mathcal{T} \in \Pi_U$ for any morphism $(U \xrightarrow{\nu} X) \in \mathcal{S}$, then $\mathcal{T} \in \Pi_X$.

Remark 2.1. Recall that a presheaf on $\mathcal{C}$ is a contravariant functor $\mathcal{C} \xrightarrow{\mathcal{G}} (\text{Sets})$. The idea is that $\mathcal{G}$ is a sheaf if and only if $\mathcal{G}(X) \rightarrow \lim_{\mathcal{S}} \mathcal{G}$ where morphisms in $\mathcal{S}$ are morphisms of X -objects, is an isomorphism for all $\mathcal{S} \in \Pi_X$.

Remark 2.2. If $\mathcal{C}$ is given by a poset, then $\mathcal{S}$ is also a poset (of $\preceq$ -predecessor of X) and

$$\lim_{\mathcal{S}} \mathcal{G} = \left\{ (g_Y)_{Y \in \mathcal{S}} \in \prod_{Y \in \mathcal{S}} \mathcal{G}(Y) \mid g_Y|_Z = g_Z \text{ where } Z \preceq Y \right\}.$$

Here $\mathcal{G}$ (the unique morphisms $Z \rightarrow Y$) are denoted by $\cdot|_Z$.

Remark 2.3. In general (fpqc-topology or proétale topology), set theoretic patterns arise which we can ? because they were $\mathcal{C}$ is given by a poset P (all all sieves the being subsets of P).

Proposition 2.1.

1. If $\mathcal{S} \subseteq \mathcal{T}$ and $\mathcal{S} \in \Pi_X$, then $\mathcal{T} \in \Pi_X$.
2. If $\mathcal{S}_1, \mathcal{S}_2 \in \Pi_X$, then $\mathcal{S}_1 \cap \mathcal{S}_2 \in \Pi_X$.

Proof. If $v \in \mathcal{S}$, then $v^* \mathcal{T} \supseteq v^* \mathcal{S} = (\text{all-sieve on the source } U \text{ of } \nu)$ hence $v^* \mathcal{T} = (\text{all sieve on } U) \in \Pi_U$ by GT-? By GT-Loz, $T \in \Pi_X$.

Let $\mathcal{T} = \mathcal{S}_1 \cap \mathcal{S}_2$. If $[U \xrightarrow{\nu} X] \in \mathcal{S}_1$, $v^* \mathcal{T} = v^* \mathcal{S}_2 \in \Pi_U$ by GT- and $\mathcal{S}_2 \in \Pi_X$. By GT- $\mathcal{T} \in \Pi_X$. $\square$

From now on, $\mathcal{C}$ will be given by some topology base $\mathcal{B}$ in a topology space X turned into a poset by the inclusion.

Definition 2.4.

$$[U_i | i \in I] := \bigcap_{\substack{W\text{-sieves } \mathcal{S} \\ \forall i \in I, U_i \in \mathcal{S}}} \mathcal{S}, \\ \{U \in \mathcal{B} \mid \forall i \in I, U \subseteq U_i\},$$

where all $U_i \in \mathcal{B}_W = \{U \in \mathcal{B} \mid U \subseteq W\}$.

Definition 2.5. A W -sieve is finitely generated if it has the form $[\Omega_1, \dots, \Omega_n]$ where all $\Omega \in \mathcal{B}_W$. If several topology bases are under consideration, we will use $[\Omega_1, \dots, \Omega_n]_{\mathcal{B}}$ for disambiguation.

Definition 2.6. We denote,

$$[U_1, \dots, U_n] = (\mathcal{O}_U\text{-sieve generated by } U_i) = \bigcup_{i=1}^n \mathcal{O}_{U_i}. \quad (2.1)$$

If we specify the topology base $\mathcal{B}$, we denote,

$$[\Omega_1, \dots, \Omega_n]_{\mathcal{B}} = (\mathcal{B}\text{-sieves generated by } \Omega_i) = \bigcup_{i=1}^n \mathcal{B}_{\Omega_i}. \quad (2.2)$$

A sieve is finitely generated if it can be generated in this way by finitely many of its elements.

$$\mathcal{S} / = U \iff \mathcal{S} \text{ is a } U\text{-sieve and } \mathcal{S} \in \Pi_U.$$

Also,

$$\mathcal{S} // U \iff \mathcal{S}|_U / = U, \quad (2.3)$$

where $\mathcal{S}|_U = \mathcal{S} \cap \mathcal{O}_U$.

Proposition 2.2. Let X be a topological space and $\mathcal{B}$ be a topology base for it. Then we have a bijection between the following two sets.

- 1). Grothendieck topologies Π on $\mathcal{B}$.

- 2). Grothendieck topologies Π on $\mathcal{O}_X$ with the additional property that $[\mathcal{B}_u] \in \Pi_U$ holds for all $U \in \mathcal{O}_U$.

The bijection is explicitly given by the following way.

- a). If $\Pi^{(\mathcal{B})}$ is given, $\mathcal{S} \in \Pi_U$ if and only if $\mathcal{S} \cap \mathcal{B}_\Omega \in \Pi_\Omega^{(\mathcal{B})}$ holds for all $\Omega \in \mathcal{B}_U$.
- b). If Π is given, $\mathcal{S} \in \Pi_\Omega^{(\mathcal{B})}$ if and only if $[\mathcal{S}] \in \Pi_\Omega$, (where $[\mathcal{S}] = [\theta \mid \theta \in \mathcal{S}]$ is the sieve on $\mathcal{O}_\Omega$ generated by $\mathcal{S}$).

Proof. Omitted. Hint : it becomes easier if it is assumed that $\Omega, \Theta \in \mathcal{B} \Rightarrow \Omega \cap \Theta \in \mathcal{B}$. $\square$

Definition 2.7. A G_+ -topological space is a T_0 -space X with a Grothendieck topology on $\mathcal{O}_X$ such that

1. $\emptyset / = \emptyset$ (implies $\mathcal{G}(\emptyset) = \{0\}$ for sheaves $\mathcal{G}$ of abelian groups).
2. If $\mathcal{S} / = U$ then $U = \bigcup_{V \in \mathcal{S}} V$.

Definition 2.8. If X is a G_+ -space then a G_+ -base for X is a subset $\mathcal{B} \subseteq \mathcal{O}_X$ such that $\mathcal{B}$ is a topology base in the ??? and $[\mathcal{B}_u] / = U$ for all $U \in \mathcal{O}_U$. We say that $\mathcal{B}$ is G_{++} if in addition,

- a). $\Omega, \Theta \in \mathcal{B} \Rightarrow \Omega \cap \Theta \in \mathcal{B}$.
- b). Membership in $\mathcal{B}$ is local in the sense that $\Omega \in \mathcal{B}, V \in \mathcal{O}_\Omega$ and $[\Theta \in \mathcal{B}_\Omega \mid U \cap \Theta \in \mathcal{B}] / = \Omega$ then $V \in \mathcal{B}$.

Definition 2.9. If X is a G_+ -space then $V \in \mathcal{O}_X$ is called G_+ -quasicompact (G_{+qc}) if for all $\mathcal{S} \in \Pi_U$, there are finitely many $(V_i)_{i=1, \dots, n}$ such that

$$[V_1, \dots, V_n] / = V.$$

(i.e. every covering sieve of U has a finitely generated covering subsieve).

Example 2.1. If X is a T_0 -space it gets the structure of a G_+ -space by

$$\Pi_U = \left\{ \text{sieves } \mathcal{S} \text{ in } U \text{ such that } U = \bigcup_{V \in \mathcal{S}} V \right\}.$$

Proposition 2.3. Let X be a T_0 -space, $\mathcal{B}$ a topology base for it such that $\Omega, \Theta \in \mathcal{B}$ implies $\Omega \cap \Theta \in \mathcal{B}$. Put

$$\Pi_U = \left\{ \text{sieves } \mathcal{S} \text{ in } U \text{ such that for all } \Omega \in \mathcal{B}_U \text{ there are } (\Theta)_{i=1, \dots, n}, n \in \mathbb{N}, \Theta_i \in \mathcal{S} \text{ such that } \Omega = \bigcup_{i=1}^n \Theta_i \right\}$$

Then X becomes a G_+ -topological space and $\mathcal{B}$ is a G_+ -base and the elements of $\mathcal{B}$ are G_{+qc} .

Proof. The axiom for a Grothendieck topology are easily verified (see sheet 5). That $\emptyset/\emptyset$ is clear from the definition.

The other parts of G_+ for X is also clear. Let $U \in \mathcal{O}_X$ and $\mathcal{S}[\mathcal{B}_U]$. If $\Omega \in \mathcal{B}_U$ then $\Omega \in \mathcal{S}$ hence $n = 1, \theta_1 = \Omega$ can be taken.

Finally let $\mathcal{S}/\Omega$. If $\Theta_1, \dots, \Theta_n$ are as in the formulation of the Proposition. Then $\mathcal{T} = [\Theta_1, \dots, \Theta_n]$ then $\mathcal{T}/\Omega$ as $\Xi \in \mathcal{B}_\Omega, \Xi_i = \Xi \cap \Omega_i \in \mathcal{B} \cap \mathcal{T}$ and

$$Xi = \bigcup_{i=1}^n \Xi_i. \quad \square$$

Remark 2.4. The assertion of the Proposition 2.3 will still hold when $\Omega, \Theta \in \mathcal{B} \Rightarrow (\exists n \in \mathbb{N} \exists (\Xi_i)_{i=1, \dots, n} \in \mathcal{B}^n \text{ s.t. } \Omega \cap \Theta = \bigcup_{i=1}^n \Xi_i)$.

Definition 2.10. This is called the G_+ -topology on X enforcing the quasicompactness of the elements of $\mathcal{B}$.

Definition 2.11. A sieve in $\mathcal{O}_X$ is called a prime sieve if $n \in \mathbb{N}$ and $\bigcap_{i=1}^n U_i \in \mathcal{S}$, (intersection in X and all $U_i \in \mathcal{O}_U$) implies that there is i such that $1 \leq i \leq n$ and $U_i \in \mathcal{S}$.

Remark 2.5. It is sufficient to require this when $n = 0, (X \notin \mathcal{S})$ and $n = 2(U \cap V \in \mathcal{S} \Rightarrow U \in \mathcal{S} \text{ or } V \in \mathcal{S})$.

Proposition 2.4. For a subset $\xi \subseteq \mathcal{O}_X$ the following conditions are equivalent.

a). ξ satisfies

i). $V \in \xi, V \in \mathcal{O}_X, U \subseteq V \Rightarrow V \in \xi$.

ii). $U, V \in \xi \Rightarrow U \cap V \in \mathcal{S}$ and $V \in \xi$.

iii). If $U \in \xi$ and $\mathcal{S} \in \Pi_U$ then $\mathcal{S} \cap \xi \neq \emptyset$.

b). $\mathcal{T} = \mathcal{O}_X \setminus \xi$ is a prime sieve containing every open subset covered by it ($U \in \mathcal{O}_X$ and $\mathcal{T}/U$ implies $U \in \mathcal{T}$).

Proof. Obvious. $\square$

Definition 2.12. We call ξ a van der Put point.

Remark 2.6. The set of topos points (referring to SGA 4.1). If the category of sheaves of sets on X (as defined later on) is in canonical bijection between van der Put points.

Definition 2.13. The set of vander Put points of X is denoted by X^* .

Proposition 2.5. Let X be a G_+ -space and $U \in \mathcal{O}_U$ be an open subset. Equip U with the G_+ -structure $\Pi^{(U)} = \Pi|_{\mathcal{O}_U}$, we then have a bijection

$$\{\text{van der Put points } \xi_U \text{ of } U\} \xrightleftharpoons[\xi_U = \xi \cap \mathcal{O}_X \leftarrow \xi]{\xi_U \mapsto \xi = \{U \in \mathcal{O}_X \mid U \cap U \in \xi_U\}} U^* = \{\xi \in X^* \mid U \in \xi\}$$

We have $(U \cap V)^* = U^* \cap V^*$.

Definition 2.14. We equip X^* with the topology for which $\{U^* \mid U \in \mathcal{O}_X\}$ is a topology base.

Proposition 2.6.

- a). If $\mathcal{S} / = U$ then $U^* = \bigcup_{V \in \mathcal{S}} V^*$. This follows from Proposition 2.4 a.iii.
- b). If $\mathcal{B}$ is a G_+ base then $\mathcal{B}^* = \{\Omega^* \mid \Omega \in \mathcal{B}\}$ is a topology base for X^* . Follows from Definition 2.8.
- c). $\xi \in X^*$ is a specialization (i.e. $\xi \in \overline{\{u\}}$ of $u \in X^*$) if and only if $\xi \subseteq u$.
- d). If $\mathcal{S}$ is a maximal element of the set of sieves on X which are not in Π_X . Then $\xi = \mathcal{O}_X \setminus \mathcal{S}$ is a closed point of X^* .

Proof. We show the proof of d). Obviously, $X \notin \mathcal{S}$ by GT-Triv. Let $U_1, U_2 \notin \mathcal{S}$ such that $U = U_1 \cap U_2 \in \mathcal{S}$. By maximality of $\mathcal{S}$, $\mathcal{S}_i = \mathcal{S} \cup \mathcal{O}_{U_i} \in \Pi_X$. But $\mathcal{S}|_V = \mathcal{S}_2|_V$ for all $V \in \mathcal{S}_1$. Then $\mathcal{S} \in \Pi_X$ by GT-Loc, a contradiction. Thus $\mathcal{S}$ is a prime sieve. if $\mathcal{S}/V$ then $\mathcal{S} \cup \mathcal{O}_X \notin \Pi_X$ (otherwise $\mathcal{S}/ = X$ by GT-Loc). Then $V \in \mathcal{S}$ by maximality of $\mathcal{S}$. By Proposition 2.4, $\xi = \mathcal{O}_X \setminus \mathcal{S}$ is a van der Put point. If ξ has a proper specialization u then $T = \mathcal{O}_X \setminus u \notin \Pi_X$ and $\mathcal{T} \supseteq \mathcal{S}$ a contradiction. $\square$

Definition 2.15. Let X be a G_+ -space. We say X has sufficiently many van der Put points if for all $U \in \mathcal{O}_X$, and every U -sieve $\mathcal{S}$, $U^* - \bigcup_{V \in \mathcal{S}} V^*$ implies $\mathcal{S}/ = U$.

Remark 2.7. Definition 2.15 can be restated by saying that the converse of a) in Proposition 2.6 also holds.

Lemma 2.1. If X has sufficiently many van der Put points, then it is G_+ -quasicompact and $\mathcal{S}$ is an $\mathcal{O}_X$ -sieve such that $X^* = \bigcup_{U \in \mathcal{S}} U^*$ then $\mathcal{S} = X$.

Proof. Assume that $\mathcal{S}$ is not covering X . We apply Zorn's lemma to $\mathcal{M}$, the poset of sieves by inclusion such that $\mathcal{T} \notin \Pi_X$. This is possible as $\mathcal{M} \neq \emptyset$ because $\mathcal{S} \in \mathcal{M}$ and if $\mathcal{L} \subseteq \mathcal{M}$ is non-empty subset linearly ordered by $\subseteq$, then $\tilde{\mathcal{S}} = \bigcup_{\mathcal{T} \in \mathcal{L}} \mathcal{T}$ is a sieve containinig $\mathcal{S}$. If $\tilde{\mathcal{S}} \in \Pi_X$, then by the quqsi-compactness of X , there are $U_1, \dots, U_n$ such that $U_i \in \tilde{\mathcal{S}}$ such that $[U_1, \dots, U_n]/ = X$. Then there is $\mathcal{T} \in \mathcal{L}$ such that all $U_i \in \mathcal{T}/ = X$ by Proposition ??, which is a contradiction to $\mathcal{T} \in \mathcal{M}$.

It follows that $\mathcal{M}$ has a $\subseteq$ -maximal element. By Proposition 2.6, $x_i = (\mathcal{O}_X \setminus \mathcal{S}_{\max}) \in X^*$. If $U \in \mathcal{S} \subseteq \mathcal{S}_{\max}$, then $\xi \notin U^*$ (otherwise $U \in \xi$, but by Proposition 2.5, $\xi \cap \mathcal{S}_{\max} \neq \emptyset$). A contradiction to our assumption that $X^* = \bigcup_{U \in \mathcal{S}} U^*$. $\square$

Proposition 2.7. If X is a G_+ -space which has a G_+ -base $\mathcal{B}$ where eelemnts are G_+ -quasicompact. Then X has sufficiently many van der Put points. If $\mathcal{B}$ is closed under finite intersection in X (including intersection over an empty set thus $X \in \mathcal{B}$), then X^* is a spectral space.

Proof. Let $U \in \mathcal{O}_X$ and $\mathcal{S}$ any U -sieve such that $U^* = \bigcup_{V \in \mathcal{S}} V^*$. Let $\Omega \in \mathcal{B}_U$ and $\mathcal{S}_\Omega = \mathcal{S}|_\Omega$. Then $\Omega^* \subseteq U^* = \bigcup_{V^*} V^*$ hence $\Omega^* = \bigcup_{V \in \mathcal{S}} \Omega^* \cap V^* = \bigcup_{V \in \mathcal{S}} (\Omega \cap V)^*$. (Note that $(V \cap W)^* = V^* \cap W^*$ by definition). Some $\Omega \cap V \in \mathcal{S}_\Omega$, it follows that $\Omega^* = \bigcup_{V \in \mathcal{S}_\Omega} V^*$. By Lemma 1.1, $\mathcal{S}/ = \Omega$. Let $\mathcal{T}$ be the sieve generated by $\mathcal{B}_U$. From GT-Transitivity and what we have seen $\mathcal{S}/ = W$ where $W \in \mathcal{T}$. By GT-Loc, $\mathcal{S}/ = U$ hence X has sufficiently many van der Put points.

For the spectrality of X^* , it is easy to see that Ω^* is quasicompact if Ω is Ω_+ -quasicompact. Then $\{\Omega^* \mid \Omega \in \mathcal{B}\}$ satisfy the requirement for the quasicompactness part of the definition of spectrality. If $Z \subseteq X^*$ is a van der Put point, then $\xi = \{U \in \mathcal{O}_X \mid U^* \cap Z \neq \emptyset\}$ is a generic point of Z . T_0 is also easy to show. $\square$

Remark 2.8.

$$\text{glim}_{X^*} U = \{U \in \mathcal{O}_X \mid U \text{ contains some } \Omega \in \mathcal{B} \text{ such that } \Omega^* \in \mathcal{F}\}$$

All Ω^* with $\Omega \in \mathcal{B}$ are closed under this.

2.2 Sheaves

Definition 2.16. A presheaf on X with values in (sets, abelian groups, rings) is a contravariant functor $\mathcal{G}$ from $\mathcal{O}_X$ to one of these categories. Then all $\mathcal{G}(U)$ are objects in these categories and we have

$$V \subseteq U \Rightarrow \mathcal{G}(U) \xrightarrow{g|_V} \mathcal{G}(V),$$

which is a morphism in the target category such that $g \in \mathcal{G}$ then $g|_U = g, (g|_V)_W = g|_W$ if $W \subseteq V \subseteq U$. If this is only given on elements of a certain topology base $\mathcal{B}$, we speak of a presheaf on $\mathcal{B}$. In all cases, if a Grothendieck topology in $\mathcal{B}$ is given, then a sheaf is a presheaf $\mathcal{G}$ satisfying the additional assumption that

$$\mathcal{G}(\Omega) \xrightarrow{\sim} \varprojlim_{\theta \in \mathcal{S}} \mathcal{G} = \{(g_\theta)_{\theta \in \mathcal{S}} \in \prod_{\theta \in \mathcal{S}} \mathcal{G}(\theta) \mid g_\theta|_\Xi = g_\Xi \text{ where } \theta \in \mathcal{S}, \Xi \in \mathcal{B}_\theta\}.$$

If the map is only injective, then

Definition 2.17.

1. A stalk of $\mathcal{G}$ at $\xi \in X^*$ is such that

$$\mathcal{G}_\xi = \varinjlim_{\Omega \in \xi} \mathcal{G}(\Omega) = \{(\Omega, g) \mid \Omega \in \xi \cap \mathcal{B}, g \in \mathcal{G}(\Omega)\} / \sim$$

where the equivalent relation is given by $(g, \Omega) \sim (r, \Theta)$ if there is $\Xi \subseteq \Omega \cap \Theta$ such that $\Xi \in \mathcal{B}$ and $g|_\Xi = r|_\Xi$.

2. A G -skyscraper sheaf of $\xi \in X^*$ is

$$([G]_\xi)(U) = \begin{cases} G, & U \in \xi, \\ \{0\} & \end{cases}$$

3. When X is a G_+ -space with sufficiently many van der Put points

$$(\mathcal{G})(U) = \left\{ \gamma \in \prod_{\xi \in V^*} \mathcal{G}_\xi \mid \mathcal{S}_\gamma / = U \right\}$$

where $\mathcal{S}_\gamma$ is the sieve of or in $\mathcal{O}_X$ generated by as in Proposition 2.2 those $\Omega \in \mathcal{B}_u$ such that there is $g \in \mathcal{G}(\Omega)$ such that γ_ξ the component of ξ of γ is the image of g under $\mathcal{G}(\Omega) \rightarrow \mathcal{G}_\xi$, for all $\xi \in \Omega^*$.

Proposition 2.8. *Let X be a G_+ -space with sufficiently many van der Put points.*

- a) *The stalk is exact on presheaves.*
- b) *$[G]_\xi$ is a sheaf and $\mathcal{G} \mapsto \mathcal{G}_\xi$ is left adjoint to $G \mapsto [G]_\xi$ on (pre)sheaves.*
- c) *if $\mathcal{G}$ is a presheaf, the morphism $\mathbb{G} \rightarrow \text{Sheaf}(\mathcal{G})$ sending $g \in \mathcal{G}(U)$ to (image of g in $\mathcal{G}_\xi)_{\xi \in U^*} \in \prod_{\xi \in U^*} \mathcal{G}_\xi$ defines an isomorphisms on stalks.*
- d) *$(\mathcal{G}_\mathcal{B}) \xrightarrow{\sim} \mathcal{G}_\xi$ for all $x \in X$.*
- e) *A morphism of presheaves defining isomorphism on stalks defines an isomorphism on sheafification.*
- f) *If $\mathcal{G}$ is a sheaf, $\mathcal{G} \rightarrow \text{Sheaf}(\mathcal{G})$ is an isomorphism. If $\mathcal{G}$ is only defined on $\mathcal{B}$, $\mathcal{G} \xrightarrow{\sim} \text{Sheaf}(\mathcal{G})|_\mathcal{B}$.*
- g) *A morphism of sheaves defining isomorphism on all stalks is an isomorphism.*
- h) *Sheafification (Presheaves on $\mathcal{B}$) $\rightarrow$ (sheaves on $\mathcal{O}_X$) is left adjoint to the functor of restriction to $\mathcal{B}$.*
- i) *One has a equivalence of category between sheaves on X (for the Grothendieck topology on X) and (ordinary) sheaves on X^* , ? a sheaf $\mathcal{G}$ on X to the sheafification of the (pre)sheaf $U^* \rightarrow \mathcal{G}(U)$ on $\{U^* \mid U \in \mathcal{O}_X\}$ (a topology base for X^*) and a sheaf $\mathcal{H}$ on X^* to $\mathcal{G}(U) \rightarrow \mathcal{H}(U^*)$.*

Proof. a)-h) are showwn in some ways as in derived sheaf theory. i) follows from these □

3 General non-archimedian topological ring

3.1 General topological rings and modules

Definition 3.1. *A topological group is a group G with a topology on the underlying set such that multiplication and inversion are continuous. That is*

$$\mu(g, h) = gh, \iota(g) = g^{-1}$$

are continuous.

Definition 3.2. A topological ring is a ring R with a topology on the underlying set such that it is a topological group under addition and

$$R \times R \ni (a, b) \mapsto ab \in R$$

is continuous. (We do not assume the continuity of $a \mapsto a^{-1}$ for $a \in R^\times$).

Definition 3.3. Let R be a topological ring. A topological R -module M is an R -module with topology on its underlying set such that $(M, +)$ is a topological group and

$$R \times M \ni (a, m) \mapsto am \in M$$

is continuous.

Proposition 3.1. For a topological group G , the following conditions are equivalent.

- a). G is T_0 .
- b). G is T_1 .
- c). G is T_2 .
- d). $\{0\}$ is closed in G .

Proof.

If $x \in \overline{\{0\}}$ then $x^{-1} \in \overline{\{0\}}$ by the continuity of inversion. Hence $1 \in \overline{\{x\}}$ by continuity of $\mu(x, \cdot)$ hence T_0 is satisfied. For a) $\Rightarrow$ d), let

$$\Delta = (\text{preimage of } \{u\} \text{ under } G \times G \xrightarrow{(g,h) \mapsto g^{-1}h} G)$$

is closed thus T_2 . □

Proposition 3.2. Every open subgroup H of a topological group G is also closed, and every group $H \subseteq H' \subseteq G$ is also clopen.

Proof. We have,

$$G \setminus H = \bigcup_{g \in G/H \setminus \{0\}} gH$$

is open therefore H is closed. Every H' subset is a union of right or left H -cosets in G which is also open. □

Definition 3.4. Let R be a topological ring and M be a topological R -module. A subset $X \subseteq M$ is bounded or R -bounded if for every neighborhood U of 0 in M , there is a neighborhood V of 0 in R such that

$$V \cdot X = \{vx \mid v \in V, x \in X\} \subseteq U.$$

Example 3.1. When R has the discrete topology, every subset is bounded (take $V = \{0\}$).

Example 3.2. This defines a boundary on X . If X_1, X_2 are bounded then so is $X_1 \cup X_2$. Every $\{m\}$ is bounded, $M \subseteq N$ and N bounded then M is also bounded.

Example 3.3. If $R = K$, equipped with an valuation $|\cdot|$ and M is an (valuative) K -vector space and K is not discrete, then a subset X of M is bounded if and only if there is $B \in [0, \infty)_{\mathbb{R}}$ such that

$$\forall x \in X, \|x\| \leq B.$$

For example, take $U = \{x \in X \mid \|x\| \leq 1\}$, $B = |x|^{-1}$ where $x \in V \setminus \{0\}$ and V is as in the definition.

Definition 3.5. For subsets $X_1, X_2 \subseteq R$, we define,

$$X_1 X_2 = \{x_1 x_2 \mid x_1 \in X_1, x_2 \in X_2\}.$$

Let $X^0 = \{1\}$. We define recursively X^n such that

$$X^{n+1} = X \cdot X^n.$$

We say that a subset $X \subseteq R$ is power bounded if

$$\bigcup_{n \in \mathbb{N}} X^n$$

is bounded. It is topologically nilpotent if for every neighborhood U of 0 in R , there is $n \in \mathbb{N}$ such that

$$\bigcup_{m=n}^{\infty} X^m \subseteq U.$$

We call an element $r \in R$ topologically nilpotent / power bounded if $\{r\}$ satisfies these properties.

Proposition 3.3. Let R be a topological ring and M be a topological R -module. We have the following statements.

- a). A finite union of bounded subsets of M is bounded.
- b). Trivially, every finite subset of M is bounded.
- c). If $X_1, X_2 \subseteq M$ are bounded then so is $X_1 + X_2 = \{x_1 + x_2 \mid x_1 \in X_1, x_2 \in X_2\}$.
- d). If $X \subseteq R$ and $Y \subseteq M$ are bounded then so is $XY \subseteq M$.

Proof.

c). Let U be a neighborhood of 0 in M , there are neighborhood $\tilde{U}$ of 0 in M such that $\tilde{U} + \tilde{U} \subseteq U$ and another neighborhoods V_1, V_2 of 0 in R such that $V_1 X_1, V_2 X_2 \subseteq \tilde{U}$. Take $V = V_1 \cap V_2$ in Definition 3.4. Proofs for the other statements are also easy. $\square$

Proposition 3.4. *If $X \subseteq M$ is R -bounded and $M \xrightarrow{f} N$ a continuous morphism of R -modules then $f(X)$ is bounded in N .*

Proof. If $U \subseteq N$ is a neighborhood of 0, let $\tilde{U} = f^{-1}(U)$ and V a neighborhood of 0 in R such that $VX \subseteq \tilde{U}$. Then $Vf(X) \subseteq U$. $\square$

It is not always true that the image of a bounded subset of a R under a morphism $R \rightarrow S$ of topological ring is bounded. (The image is R -bounded but not necessarily S -bounded) for example take $R = \mathbb{Q}_p, S = \mathbb{Q}_q$. However, of the following conditions on R asserts that this does not happen.

R has a topologically nilpotent unit (T)

R has a sequence of units converging to 0 (H)

Every neighborhood of 0 intersects $R^\times$ (E)

Proposition 3.5. *Let $R \xrightarrow{f} S$ be a continuous morphism of topological ring. If R has (T),(H),(E), then so has S and the image $f(X)$ of a bounded subset of R is a bounded subset of S .*

Proof. Rather easy. if U is a neighbourhood of 0 in S and R has (E), put $\tilde{U} = f^{-1}(U)$ and $r \in R^\times \cap \tilde{U}$. $f(r) \in S^\times \cap U$. If

$$\lim_{j \rightarrow \infty} r_j = 0$$

in R and all $r_j \in R^\times$ (where $r_j = r^j$ and $r \in R^\times$)

$$\lim_{j \rightarrow \infty} f(r_j) = 0$$

in S and all $f(r_j) \in S^\times$. (respectively $f(r_j) = s^j$ with $s = f(r) \in S^\times$) showing that (T) and (H) are likewise inherited by S . If R has (E) and $X \subseteq R$ bounded and U is a neighborhood of 0 in S , there is $\tilde{U}$ a neighborhood of 0 in R such that $\tilde{U}\tilde{U} \subseteq U$. Let $U' = f^{-1}(\tilde{U}) \subseteq R$ a neighborhood of 0. There is a neighbourhood V' of 0 in R such that $V'X \subseteq U'$. As R has (E), we can take $\rho \in R^\times \cap f^{-1}(\tilde{U})$. Take $V = \rho V'$ then $Vf(X) \subseteq U$. $\square$

Proposition 3.6. *Let R be a topological ring.*

- a). *Every power bounded subset is bounded.*
- b). *A finite union of power bounded subsets is power bounded.*
- c). *A subset $X \subseteq R$ which is both bounded and topologically nilpotent is power bounded.*
- d). *A finite union of subsets which are both bounded and topologically nilpotent is topologically nilpotent.*
- e). *If X is power bounded or topologically nilpotent and Y is topologically nilpotent, XY is then topologically nilpotent.*

f). The image of a topologically nilpotent $X \subseteq R$ under a morphism $R \rightarrow S$ of topological rings is topologically nilpotent.

g). If R has (E), $R \xrightarrow{f} S$ is a morphism of topological rings and $X \subseteq R$ is bounded in R . Then $f(X)$ is bounded in S .

Proof. Mostly straightforward. $\square$

Definition 3.6. Let R be a topological ring, we define the following subsets.

$$R^\circ = \{r \in R \mid r \text{ is power bounded}\}, R^{\circ\circ} = \{r \in R \mid r \text{ is topologically nilpotent}\}.$$

Remark 3.1. Without further conditions, these may fail to be additionally closed. By Proposition 3.3 b), and Proposition 3.6 a), $R^{\circ\circ} \subseteq R^\circ$ always holds.

Definition 3.7. Let $(m_n)_{n \in \mathbb{N}}$ be a sequence in M . It is Cauchy if for every neighborhood U of 0 in M , there is k such that $m_i - m_j \in U$ where $\min\{i, j\} \geq k$. M is sequentially complete if every Cauchy sequence has a limit in M .

Definition 3.8. A filter $\mathcal{F}$ of M is Cauchy if every neighborhood U of 0 in M , there is $X \in \mathcal{F}$ such that $x - y \in U$ where $x, y \in X$. M is complete if every Cauchy filter has a limit.

Definition 3.9. The sequential completion of M is a set of equivalence classes of Cauchy sequences in M where two Cauchy sequences $(m_i), (n_i)$ are equivalent if for every neighborhood U of M , there is $k \geq 0$ such that $i \geq k$ then $m_i - n_i \in U$.

Remark 3.2. If G is a non-commutative topological group, then it typically requires that both $g_i g_j^{-1} \in U$ and $g_i^{-1} g_j \in U$ where $\min\{i, j\} \geq k$. Respectively, $x^{-1}y \in U$ and $xy^{-1} \in U$ for $x, y \in X$ for filters.

Remark 3.3. One can complete by taking equivalence classes of Cauchy filters. But how to continue multiplication in that completion of R^2 . (Sequential completion does not have this problem).

Remark 3.4. It is known that topological vector space over $\mathbb{C}$ or $\mathbb{R}$ is the quotient of a complete topological vector space V over that field such that every bounded subset of V is contained in a finite dimensional subspace.

Proposition 3.7. Every quotient of a complete topological group with a countable neighborhood base of 1 by a closed normal divisor is complete.

3.2 Continuous Valuations

For an ordered group Γ equip $\Gamma \cup \{0\}$ with the topology where U is open if and only if $0 \notin U$ or there is $\gamma \in \Gamma$ such that $(0, \gamma)_\Gamma \subseteq U$.

Consider a valuation $A \xrightarrow{v} \Gamma$ where A is a topological ring. Recall that Γ_v is the convex subgroup of Γ such that

$$\Gamma_v = \langle \{v(a) \mid a \in A \cdot \text{supp } v\} \rangle_\Gamma.$$

Also recall that for $a \in A$ and $\gamma \in \Gamma$,

$$K_{v < a} = \{\alpha \in A \mid v(\alpha) < v(a)\}, K_{v < \gamma} = \{\alpha \in A \mid v(\alpha) < \gamma\}.$$

Similarly,

$$K_{v \leq a} = \{\alpha \in A \mid v(\alpha) \leq v(a)\}, K_{v \leq \gamma} = \{\alpha \in A \mid v(\alpha) \leq \gamma\}.$$

They come with an exception such that

$$K_{v < a} = \emptyset,$$

when $a \in \text{supp } \gamma v$. They are subgroups of $(A, +)$. Note that a subgroup $H \subseteq G$ of a topological group G is open if and only if it is a neighbourhood of 0. Also recall that subgroups containing open subgroups are also open.

Proposition 3.8. *The following conditions are equivalent.*

- a). $A \xrightarrow{v} \Gamma_v \cup \{0\}$ is continuous.
- b). For all element γ of the convex hull of Γ_v in Γ , $K_{v < \gamma}$ is open.
- c). For all $\gamma \in \Gamma_v$, $K_{r < \gamma}$ is open.
- d). For all $a \in A \setminus \text{supp}(v)$, $K_{r < a}$ is open.
- e). For all $a \in A \setminus \text{supp}(v)$, $K_{r < a}$ is a neighbourhood of 0.

Proof. By the previous argument d) and e) are equivalent. d) $\Rightarrow$ c) is trivial. For c) $\Rightarrow$ a), consider $\gamma \in \Gamma_v \cup \{0\}$ and a neighbourhood U of γ in $\Gamma_v \cup \{0\}$. We must show that $v^{-1}(U)$ is a neighbourhood of a in A . If $v(a) \neq 0$, it is sufficient to consider

$$U = \{v(a)\}.$$

Then,

$$v(b) = v(a)$$

where $b \in K_{r < a} + a$, and the set of such b is a neighborhood of a in A . If $v(a) = 0$, we may assume $U = [0, \gamma)_{\Gamma_v \cup \{0\}}$ with $\gamma \in \Gamma_v$. Then $v^{-1}(U) = K_{v < \gamma}$ is open by the assumption c).

a) $\Rightarrow$ d) is trivial by the definition of the topology on $\Gamma_v \cup \{0\}$, $K_{v < a}$ being $v^{-1}([0, v(a))_{\Gamma_v \cup \{0\}})$ where

$$[0, v(a))_{\Gamma_v \cup \{0\}}$$

is a neighbourhood of 0 in $\Gamma_v \cup \{0\}$.

d) $\Rightarrow$ b), every $\gamma \in \Gamma_v$ has the form,

$$\gamma = \frac{v(a)}{v(b)},$$

with $a, b \in A \setminus \text{supp } v$. Then $K_{v < \gamma}$ is the inverse image under the endomorphism $A \xrightarrow{\cdot b} A$ of the topological group $(A, +)$ of the open subgroup $K_{r < a}$. Hence it is also an open subgroup. $\square$

Definition 3.10. A valuation of A is called continuous if the equivalent conditions in Proposition 3.8 hold.

Remark 3.5. 1. This only depends on the equivalent class of v .

2. $\Gamma_v \cup \{0\}$ carries the topology induced from $\Gamma \cup \{0\}$ only if Γ is the convex hull of Γ_v in Γ . Hence Γ can be replaced by Γ_v in c) only if Γ is equal to that convex hull (in which case the c) is b).

Example 3.4. 1. $\mathbb{Q}_p \xrightarrow{|\cdot|_p} [0, \infty)_{\mathbb{R}}$ is a continuous valuation.

2. The trivial valuation for $\mathfrak{p} \in \text{Spec } A$ is continuous if and only if $\mathfrak{p}$ is open.
3. If $\mathfrak{p}$ is an open prime ideal, then any valuation with support $\mathfrak{p}$ is open.
4. For $r \in (0, 1]_{\mathbb{R}}$, the valuation N_r from 1.1(1) (need to look for the reference) is continuous.

Proposition 3.9. If v is a continuous valuation. Then $\text{supp } v$ is closed.

Proof. Indeed $\text{supp } v = v^{-1}(\{0\})$ and $\{0\}$ is closed in $\Gamma_v \cup \{0\}$. $\square$

Definition 3.11. We denote the subset equivalence classes of continuous valuations of A to be $\text{Cont}(A) \subseteq \text{Spv}(A)$. Let $A^+ \subseteq A$ be a subset we then denote $\text{Spa}(A, A^+)$ to be the set of equivalence classes of continuous valuations v of A such that $a \in A^+ \Rightarrow v(a) \leq 1$.

Remark 3.6.

1. At this point, we do not yet introduce a topology on these sets.
2. If $A = \mathbb{T}_1$ as in 1.1.(1)(look at the first part and take the reference) and $A^+ = A^\circ$. $N_r \in \text{Spa}(A, A^+)$ if and only if $r \in (0, 1)_{\mathbb{R}}$.

Definition 3.12. A Oka system is a ring A and a set $\mathfrak{M}$ of ideals in A such that for any ideal I in A , $a \in A$ and the ideal $I + aA$ and

$$(I : a) = \{\alpha \in A \mid a\alpha \in I\}$$

both belong to $\mathfrak{M}$ then $I \in \mathfrak{M}$.

Example 3.5. The set of finitely generated ideals in A in Oka topology in Proposition 1.23 as this Oka system gives Eitoi-Nagata?. If all prime ideals of A are finitely generated, A is Noetherian.

Proposition 3.10. Let $\mathfrak{M}$ be an Oka system and $\mathfrak{p} \neq A$ is a $\subseteq$ -maximal element of the set of ideals of A not belonging to $\mathfrak{M}$. Then $\mathfrak{p} \in \text{Spec } A$.

Proof. Assume $ab \in \mathfrak{p}$. Let $a \notin \mathfrak{p}, b \notin \mathfrak{p}$.

$$\mathfrak{p} + aA \text{ and } (\mathfrak{p} : a)$$

contain $\mathfrak{p}$ and not equal to $\mathfrak{p}$ as $b \in (\mathfrak{p} : a)$. By the maximality, both belong to $\mathfrak{M}$ thus $\mathfrak{p} \in \mathfrak{M}$ which is a contradiction. $\square$

Proposition 3.11. *Let $\mathfrak{M}$ be a set of ideals such that $I \subseteq J$ and $I \in \mathfrak{M}$ implies $J \in \mathfrak{M}$. Suppose further that $\mathfrak{M}$ is closed under products, that is $I, J \in \mathfrak{M}$ then $I \cdot J \in \mathfrak{M}$. Then $\mathfrak{M}$ is an Oka system.*

Proof. Indeed $I \supseteq (I + aA)(I : a)$ holds. Therefore by definition, it is Oka. $\square$

Remark 3.7. *The Oka system from Example 3.5 satisfies the assumptions if and only if A is Noetherian.*

Example 3.6. *The set $\mathfrak{M}_S$ of ideals not not disjoint form a multiplicative subset $S \subseteq A$ is an Oka system. Then every $\subseteq$ -maximal non-element of $\mathfrak{M}_S$ is a prime ideal. (This could also be seen by assuming that all maximal ideals of A_S are prime). By Zorn's lemma, $\mathfrak{M}_S$ is equal to the set of all ideals (i.e. $0 \notin S$), there is a prime ideal $\mathfrak{p} \in \mathfrak{M}_S \cap \text{Spec } A$.*

Proposition 3.12. *Let v be a valuation of a topological ring A such that $K_{r \leq 1}$ is a neighbourhood of 0 in A and such that for each $a \in A \setminus \text{supp } v$, there is $b \in A$ such that $v(ab) > 1$. Then v is continuous.*

Proof. If $a \in A \setminus \text{supp } v$, and b is as above, then $K_{v < a}$ contains the inverse image under the continuous endomorphism, $A \xrightarrow{b} A$ of $K_{r \leq 1}$ which is a neighbourhood of 0. Then Proposition 3.8 e) is satisfied. $\square$

The following conditions [Hub 93a, Lemma 3.26],[Hub 93b, Lemma 3.3.(i)] or Prop III.1.4(i)/III.4.4.(i) in Morel's text.

Proposition 3.13. *Let A be a topological ring, A^+ be an open subring of A which is integrally closed in A and $\alpha \in A \setminus A^+$. Then there is $v \in \text{Spa}(A, A^+)$ such that $v(\alpha) > 1$.*

Proof. By assumption, α is not integral over A^+ , hence $S = \{p(\alpha) | p \in A^+[T] \text{ monic}\}$ is a multiplicative subset of A not containing 0. By Example 3.6, there is $\mathfrak{p} \in \text{Spec } A$ maximal among all ideals of A disjoint from S , and this is a prime ideal. It follows from the maximality of $\mathfrak{p}$, that for every $a \in A \setminus \mathfrak{p}$, there is $b \in A$ and a monic polynomial $P \in A^+[T]$ such that $P(\alpha) - ab \in \mathfrak{p}$. Since $\mathfrak{p}$ is disjoint from S , the image $\alpha \bmod \mathfrak{p}$ of α in $k(\mathfrak{p})$ is not integral over the image $A^\#$ of A^+ in $k(\mathfrak{p})$. By Proposition 1.4.1, this is a valuation ring R of $k(\mathfrak{p})$ such that $A^\# \subseteq R$ and $\alpha \notin R$. Consider v , the composition,

$$A \rightarrow k(\mathfrak{p}) \xrightarrow{\text{valuation defined by } R \text{ Prop 1.4.1}} \Gamma.$$

Since $\alpha \bmod \mathfrak{p} \notin R$, $v(\alpha) > 1$. By the definition, $K_{v \leq 1}$ contains A^+ as A^+ and $\mathfrak{p} \subseteq A^\# \subseteq R$, hence $K_{v \leq 1}$ is open. If $a \in A$ and $v(a) \neq 0$ then $a \notin \mathfrak{p}$ and we have $b \in A$ and a monic $P \in A^+[T]$ such that $P(\alpha) - ab \in \mathfrak{p}$. Since P is monic and $v(A^+) \subseteq [0, 1]_{\Gamma \cup \{0\}}$, $P(\alpha) > 1$. Since $\text{supp } v = \mathfrak{p}$, $v(ab) = v(P(\alpha)) > 1$. By Proposition 3.12, v is continuous. $\square$

Corollary 3.1. *For a subset A^+ of A , the following conditions are equivalent.*

a). $A^+ = \{a \in A \mid |a|_x \leq 1 \text{ for all } x \in \text{Spa}(A, A^+)\}$

b). A^+ is the intersection of a set of open and integrally closed subrings of A .

Proof. For $b) \Rightarrow a)$, note that $\subseteq$ in a) is trivial and $\supseteq$ is by Proposition 3.13. For $a) \Rightarrow b)$, note that $K_{x \leq 1}$ for $x \in \text{Spa}(A, A^+)$ are integrally closed (by Fact 1.3.5) and open (by Proposition 3.8 a)) subrings of A . $\square$

References

- [1] Huber, R *Continuous valuations*, Mathematische Zeitschrift